

End-to-End Windows 10 → Windows 11 Migration (Portfolio Lab Guide)

Project Type: Hands-on Lab & Portfolio Project

Role Simulated: IT Support / Systems Administrator

Scenario: Prepare a Windows 10 system that meets Windows 11 requirements and perform a supported in-place upgrade, following enterprise-style migration practices.

1. Project Objective

The objective of this lab is to:

- Build a Windows 10 virtual machine that is eligible for Windows 11
- Validate hardware and firmware requirements
- Plan backup and rollback options
- Perform a supported in-place upgrade to Windows 11
- Verify system stability and user impact post-migration

This walkthrough mirrors how real IT teams handle Windows feature upgrades in pilot or pre-production environments.

2. Tools & Requirements

2.1 Host System

- Windows 11
- Minimum 16 GB RAM recommended (8 GB minimum)
- At least 80 GB free disk space, preferably 100 GB

2.2 Virtualization Platform

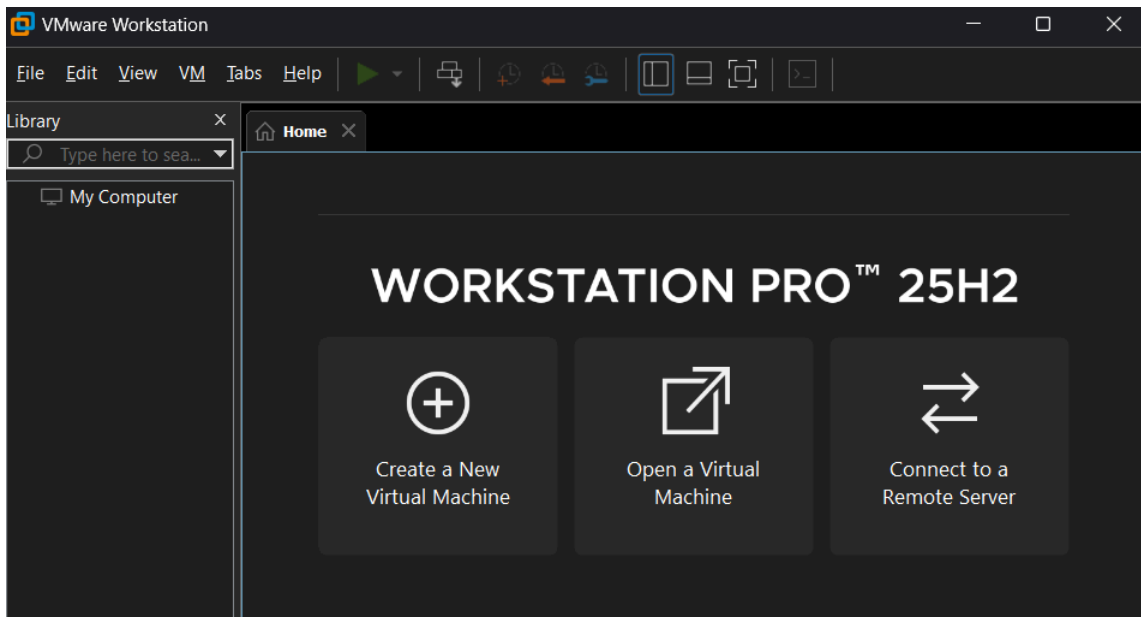
- VMware Workstation Pro / Player ([Download Here](#))

VMware is recommended because it natively supports **UEFI**, **Secure Boot**, and **TPM 2.0**, which are mandatory for Windows 11.

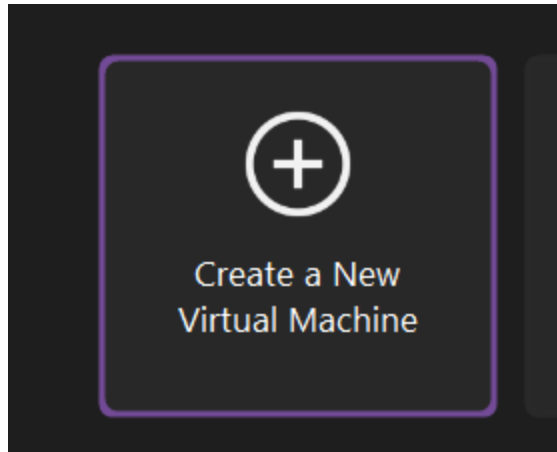
3. Phase 1 – Create a Windows 10 VM That Is Windows 11–Eligible

3.1 Create a New Virtual Machine

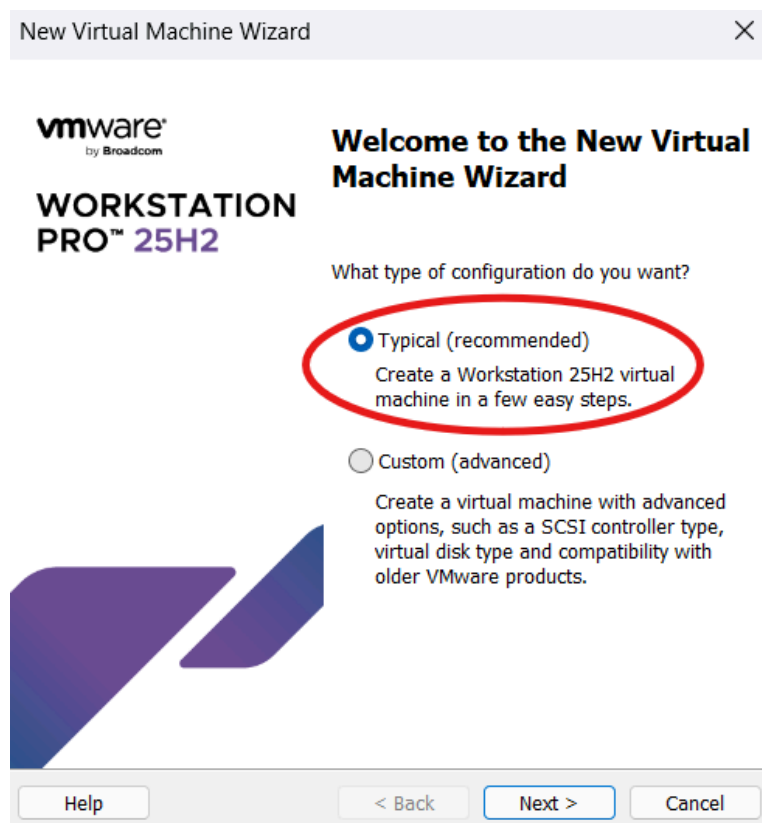
1. Open VMware Workstation Pro



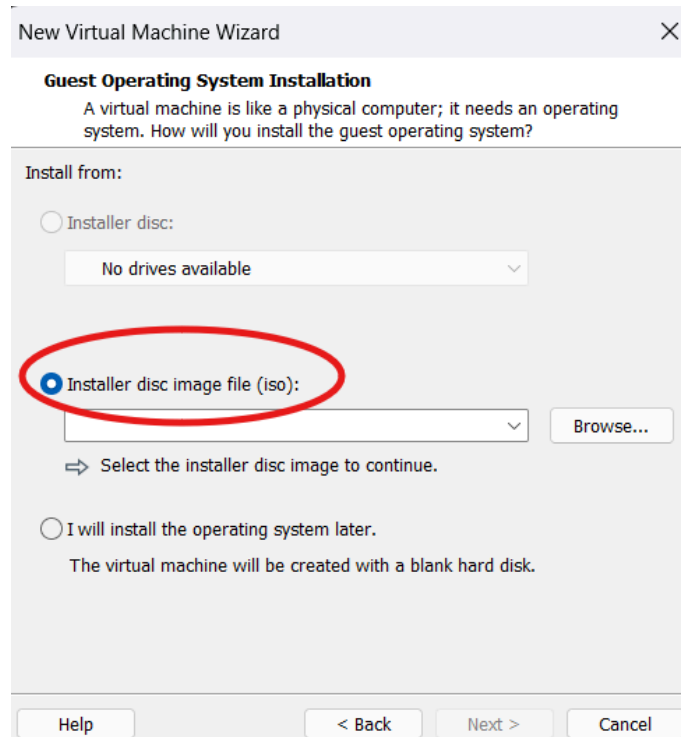
2. Select Create a New Virtual Machine



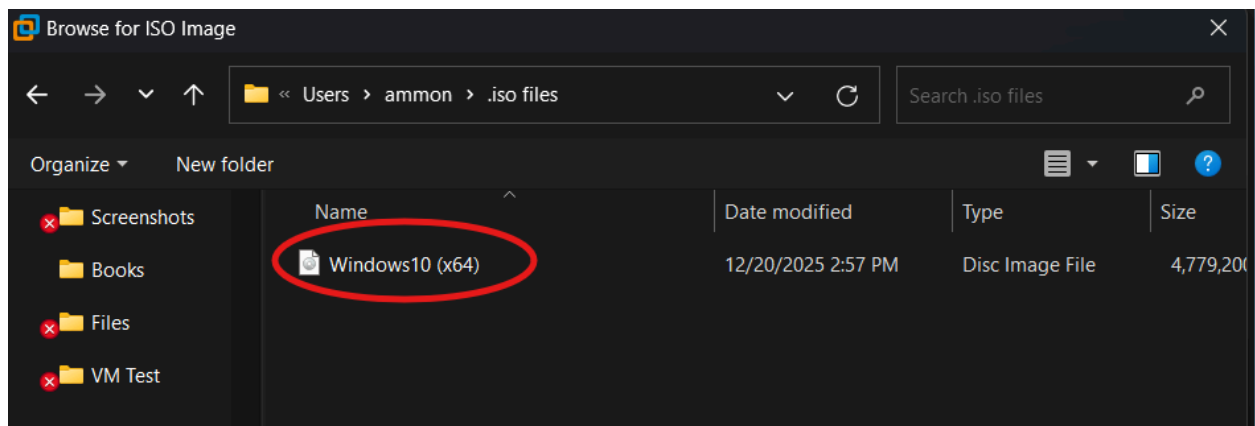
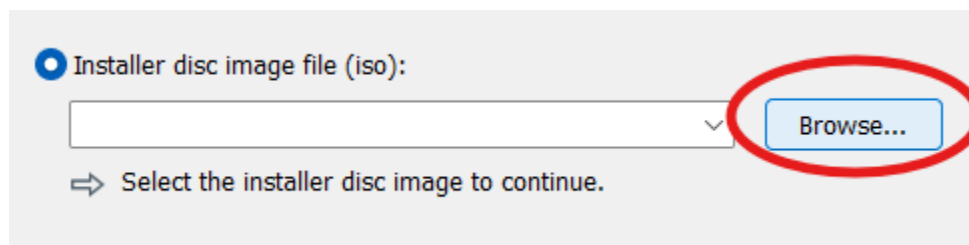
3. Choose **Typical (Recommended)**



4. Select **Installer disc image file (ISO)**



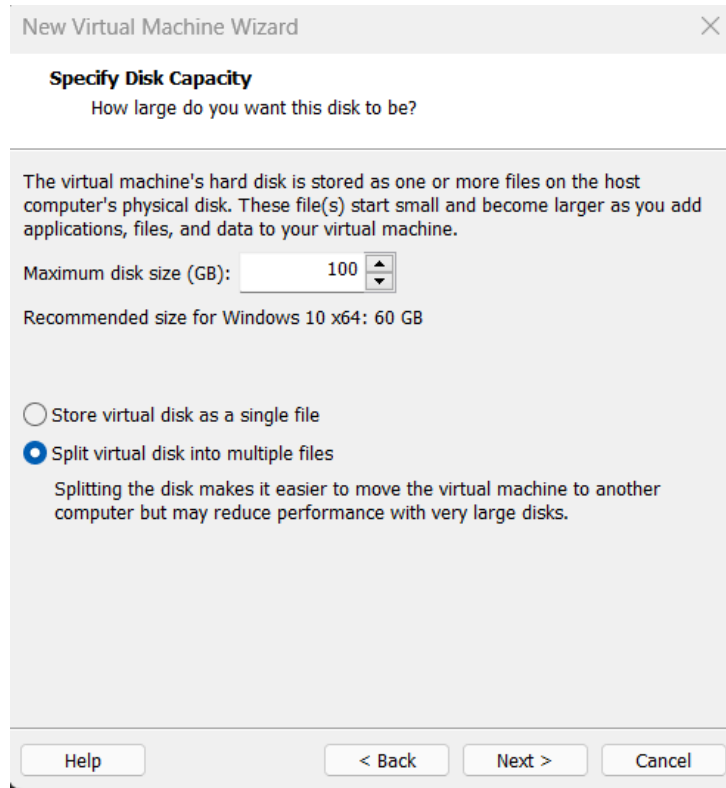
5. Browse to your **Windows 10 64-bit ISO**



3.3 Disk & Hardware Settings

Configure the VM with Windows 11 requirements in mind:

- **Disk Size:** 100 GB (Need to set it above the recommended 60GB to avoid any kind of upgrade failures)
- Select **Split virtual disk into multiple files**



- **Memory:** 8 GB RAM (4 GB minimum)

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 8192 MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB

Recommended memory
2 GB

Guest OS recommended minimum
2 GB

- Number of processors: 1
- Number of cores per processor: 2

Processors

Number of processors: 1

Number of cores per processor: 2

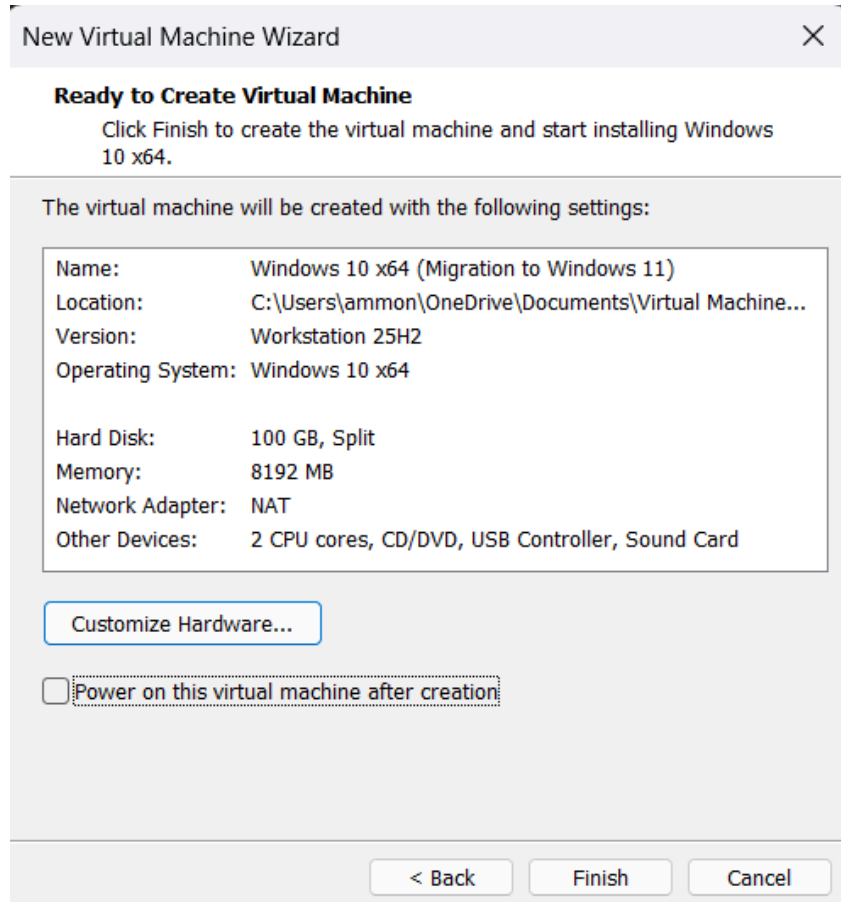
Total processor cores: 2

Virtualization engine

☐ Virtualize Intel VT-x/EPT or AMD-V/RVI

☐ Virtualize CPU performance counters

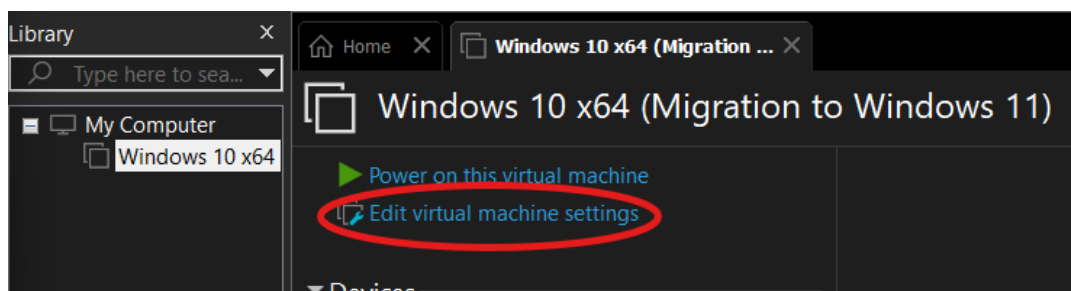
☐ Virtualize IOMMU (IO memory management unit)



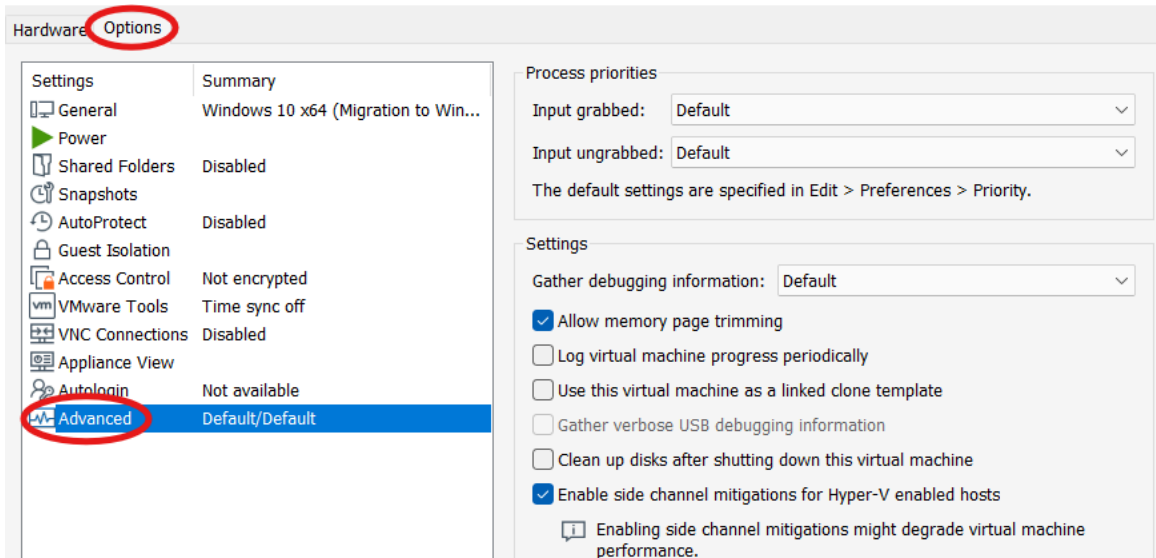
Do **not** power on the VM yet. Click **Finish**.

3.3 Enable UEFI Firmware

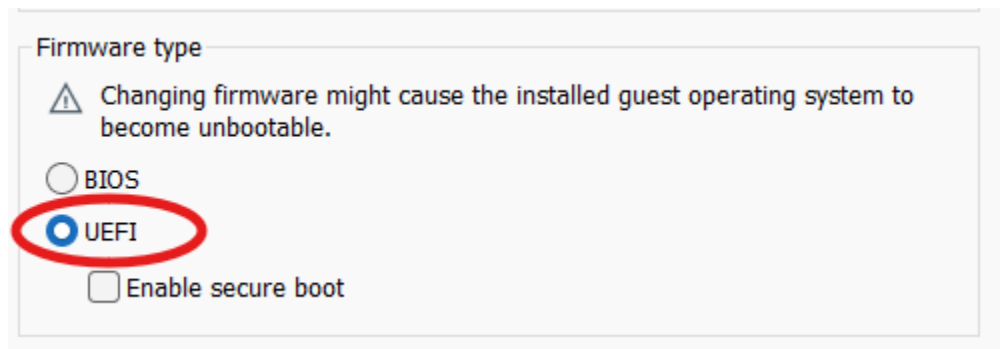
1. Select the VM → **Edit virtual machine settings**



2. Go to **Options** → **Advanced**



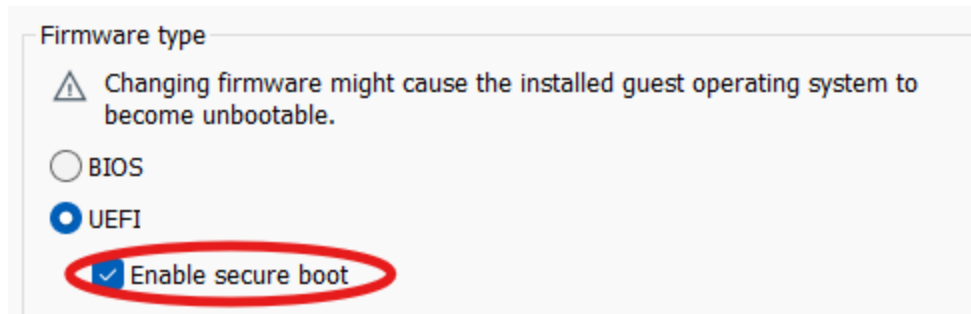
3. Set Firmware type to UEFI



This is mandatory for Secure Boot and Windows 11.

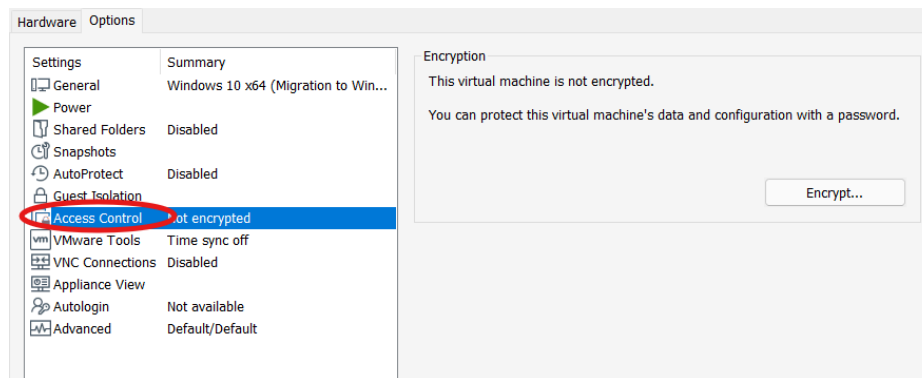
3.4 Enable Secure Boot

1. Still under **Options** → **Advanced**
2. Enable **Secure Boot**

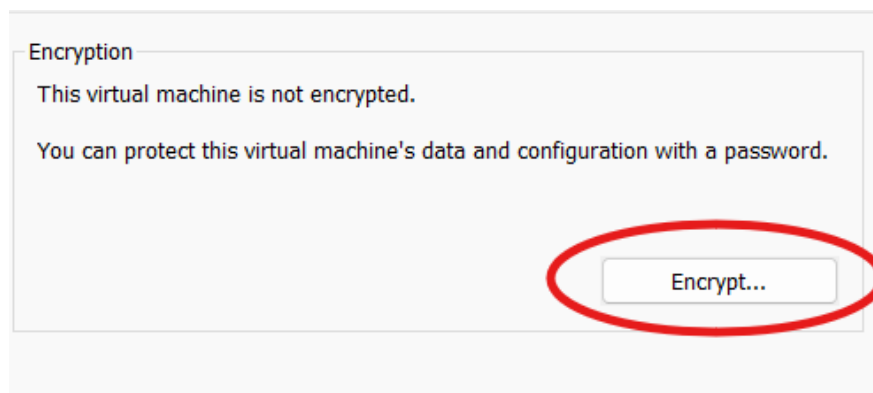


3.5 Encrypt the Virtual Machine

1. Still under **Options** → **Access Control**



2. Click **Encrypt**



3. Under **What to encrypt**, enable **All the files for this vm**

What to encrypt

☒ All the files (.vmdk, .vmx, etc.) for this virtual machine

☐ Only the files needed to support a virtual TPM (.nvram, .vmss, .vmem, .vmx, .vmsn)

4. Set an encryption Password

Encrypt Virtual Machine

Set the encryption type and password for this virtual machine.

What to encrypt

☒ All the files (.vmdk, .vmx, etc.) for this virtual machine

☐ Only the files needed to support a virtual TPM (.nvram, .vmss, .vmem, .vmx, .vmsn)

Password:

Confirm:

☒ Remember the password for this virtual machine in Credential Manager

⚠ Make sure to write down this password in a secure location. You will not be able to use this virtual machine without it.

Depending on the size of this virtual machine, the encryption process could take from a few minutes to a few hours.

Encrypt Cancel

What to encrypt

☒ All the files (.vmdk, .vmx, etc.) for this virtual machine

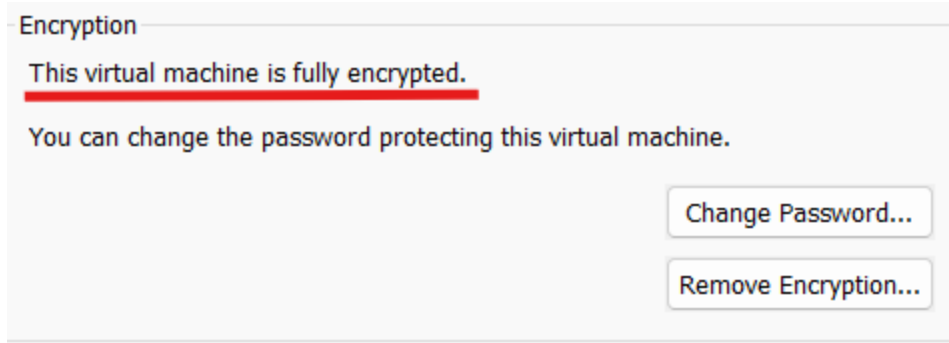
☐ VMware Workstation

Encrypting virtual machine...

Cancel

Remember the password for this virtual machine in Credential Manager

Make sure to write down this password in a secure location. You will not be able to use the virtual machine without it.

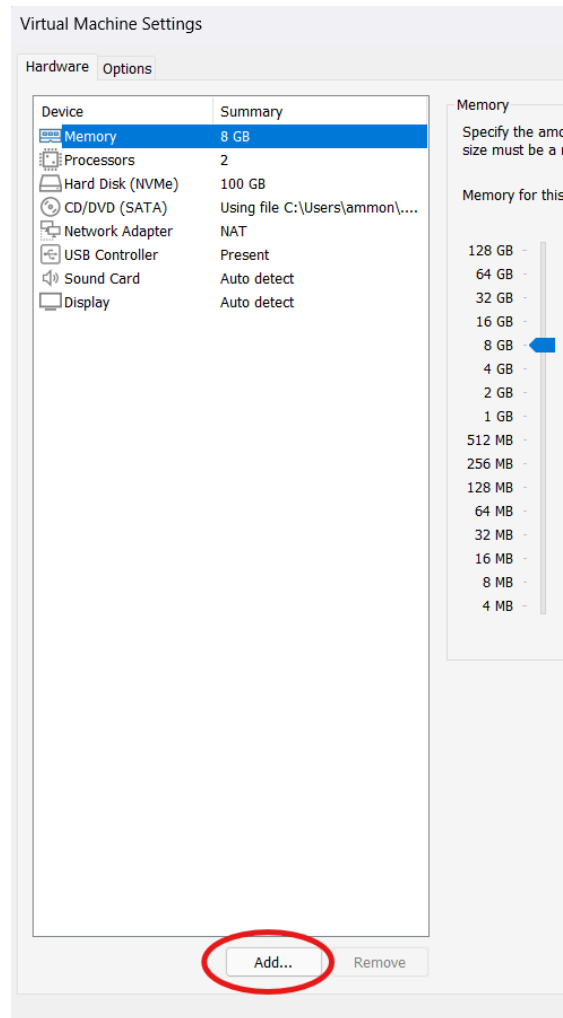


Store this password safely. You need it to modify VM hardware settings in the future.

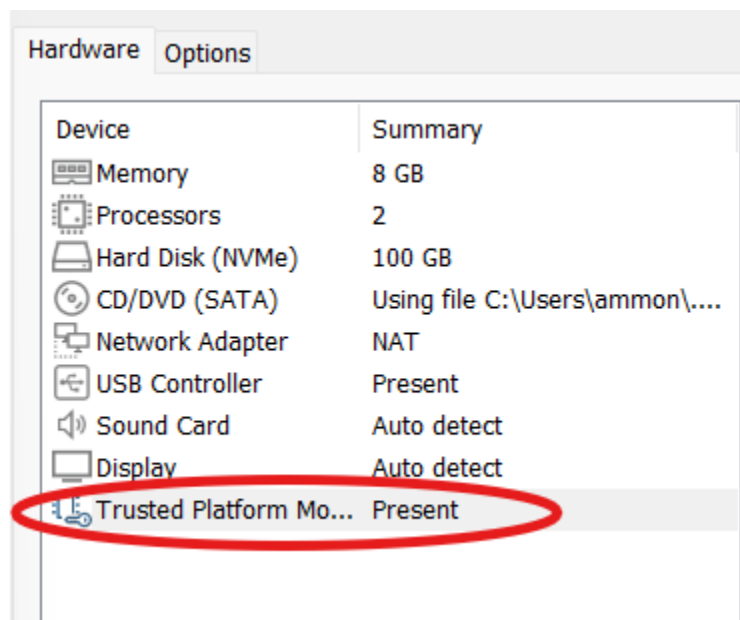
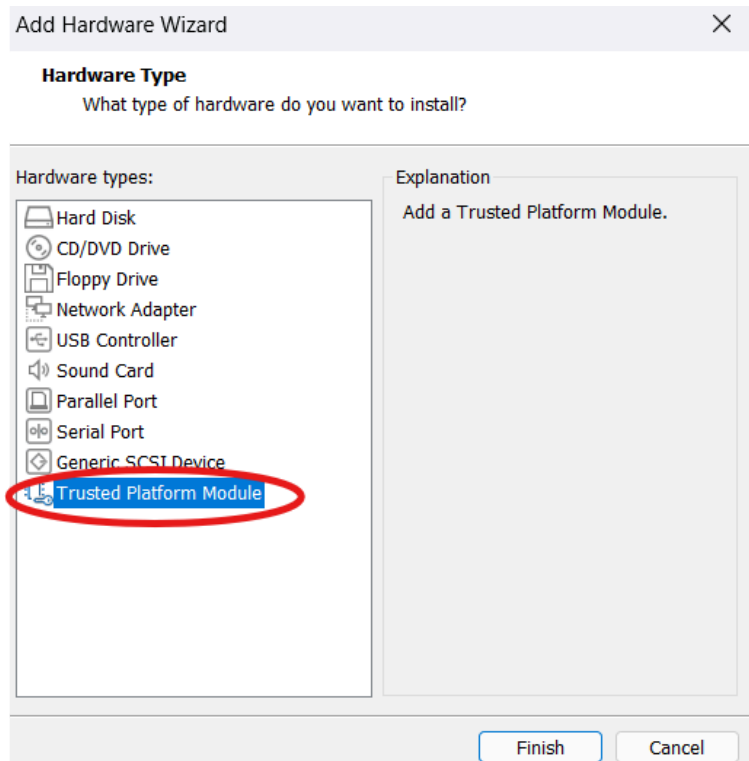
3.6 Add TPM

TPM is important for security and because Windows 11 is designed to use hardware backed security. It protects the system in ways software alone cannot.

1. In **VM Settings**, under **Hardware**, select **Add**



2. Choose **Trusted Platform Module**

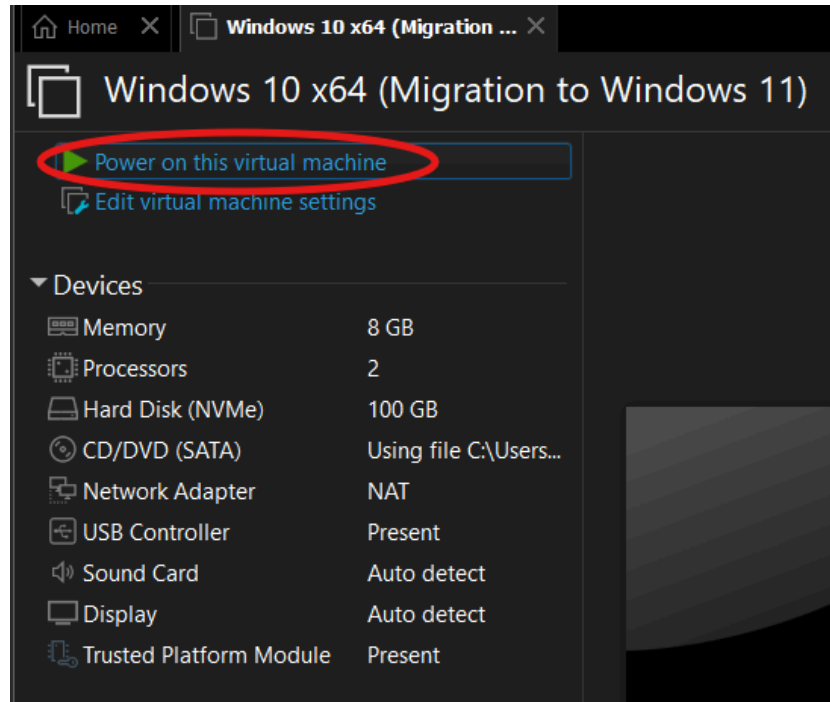


TPM 2.0 is now enabled.

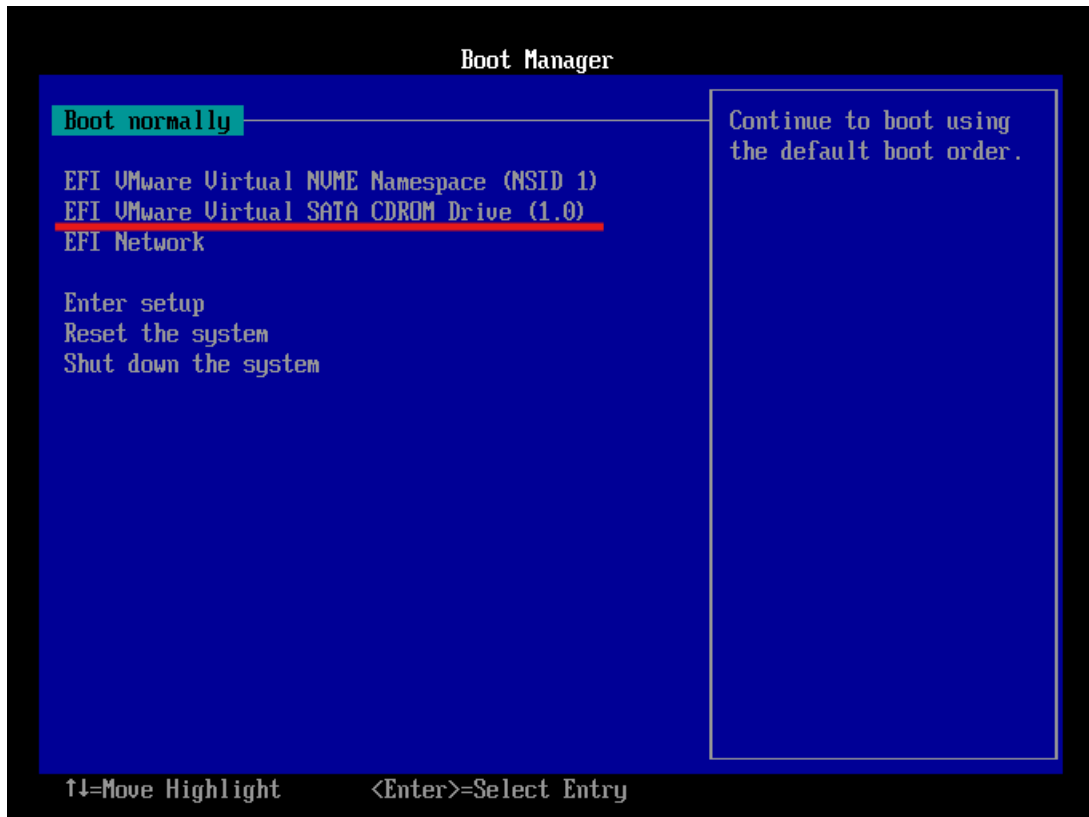
4. Phase 2 – Install Windows 10

4.1 Power On and Install

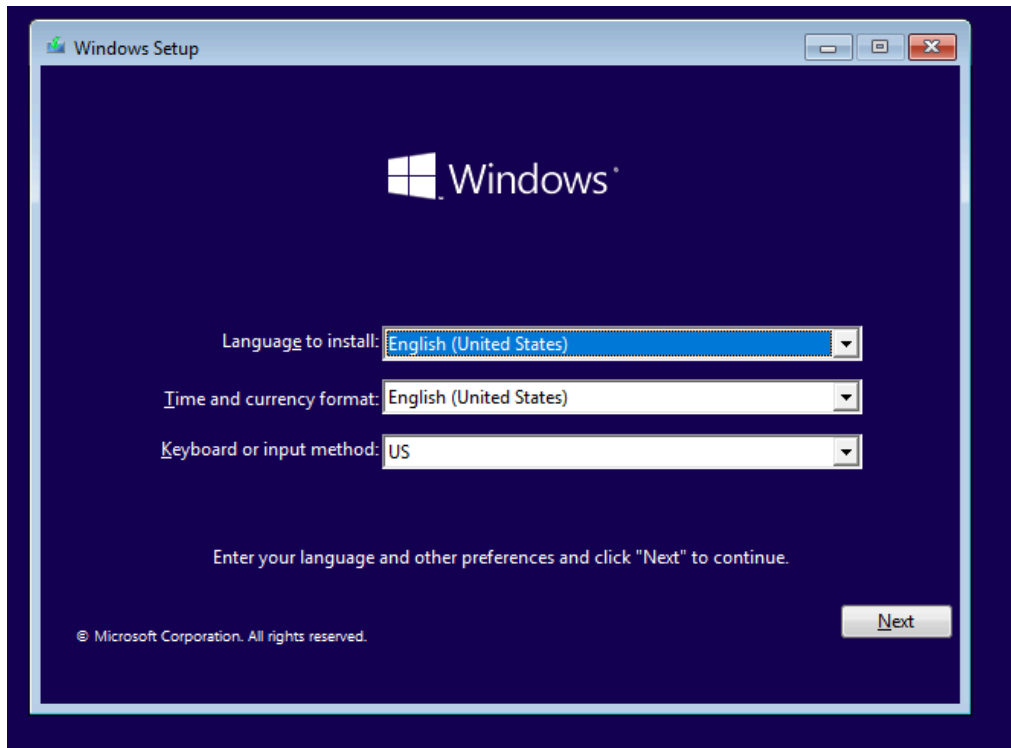
1. Start the VM



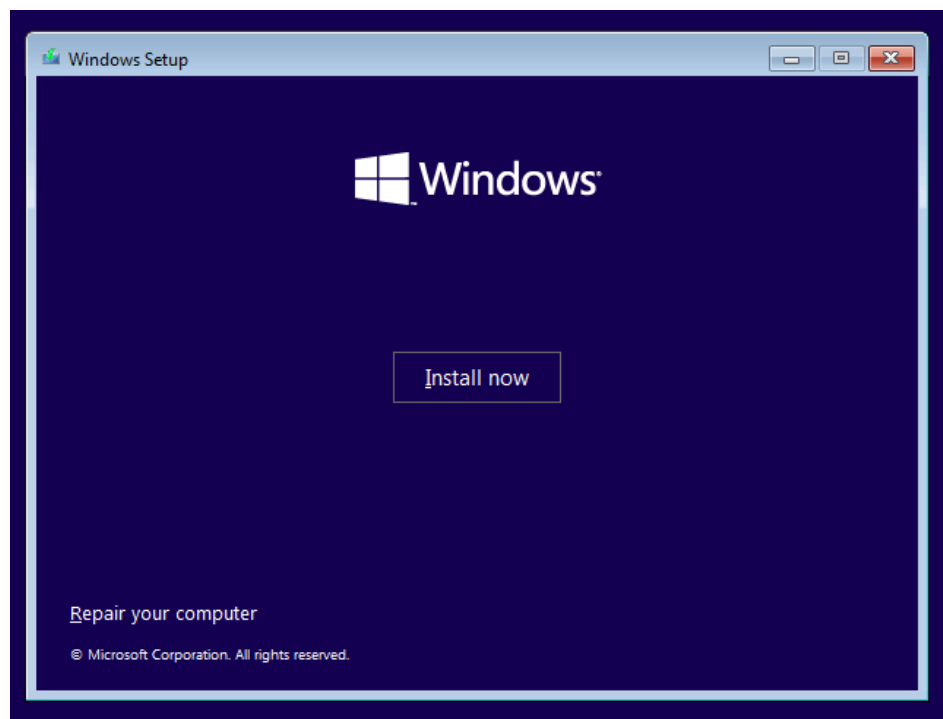
2. Once taken to **Boot Manager**, select **EFI VMware Virtual SATA CDROM Drive** (this is the Windows 10 iso)



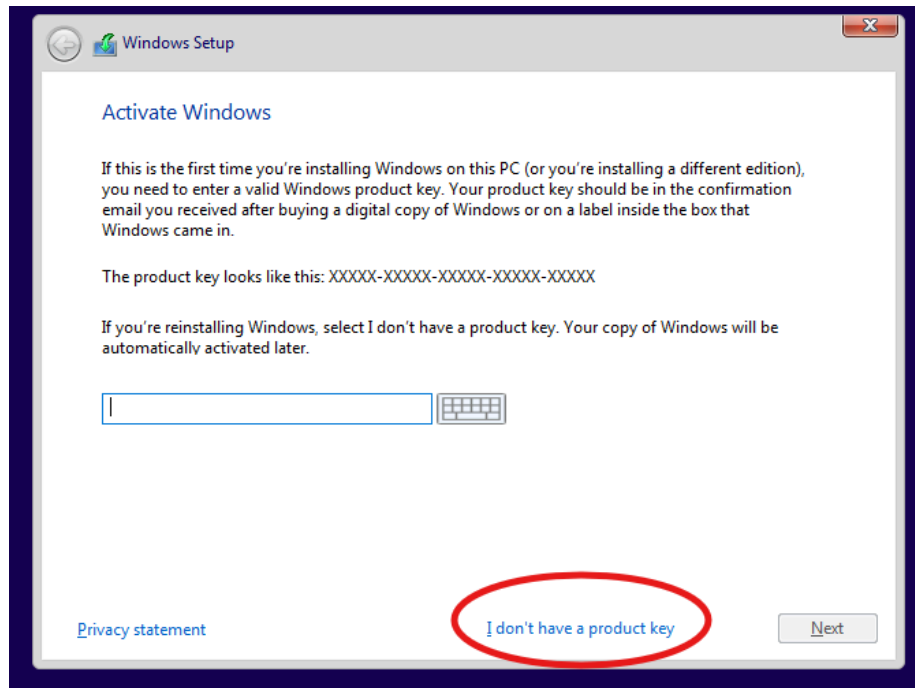
3. Follow Windows 10 installation steps:
- Language & region selection



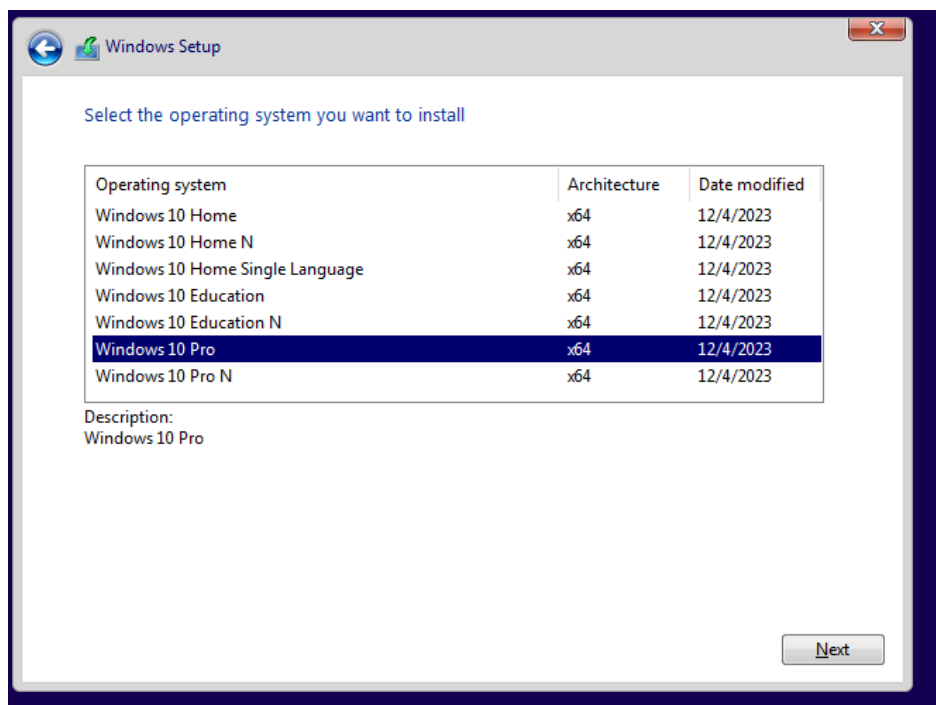
- Install Windows 10



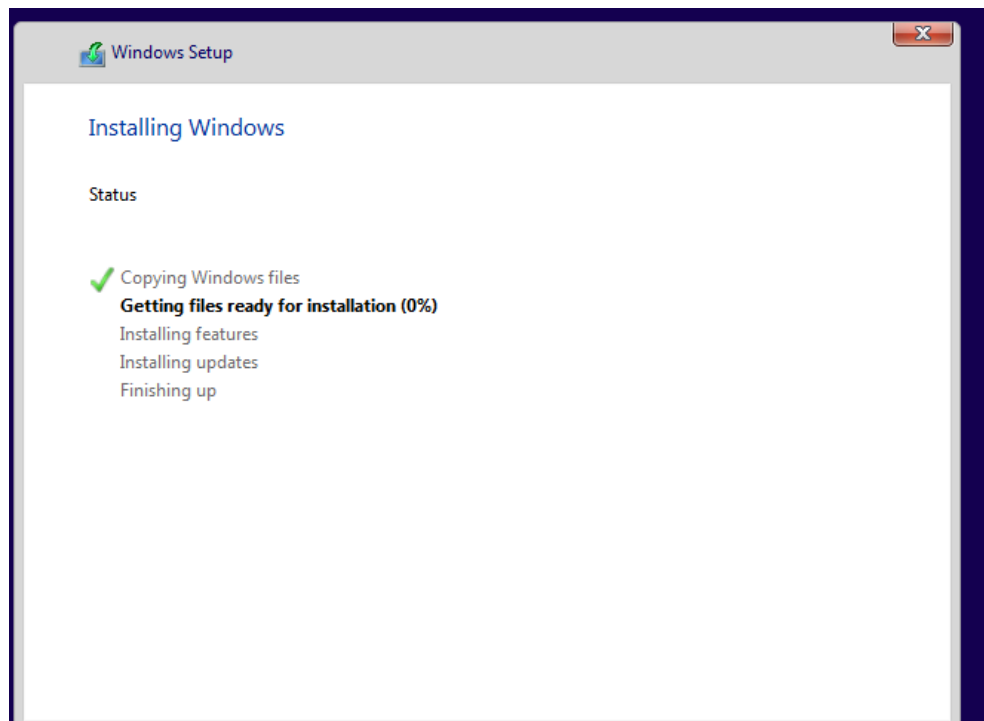
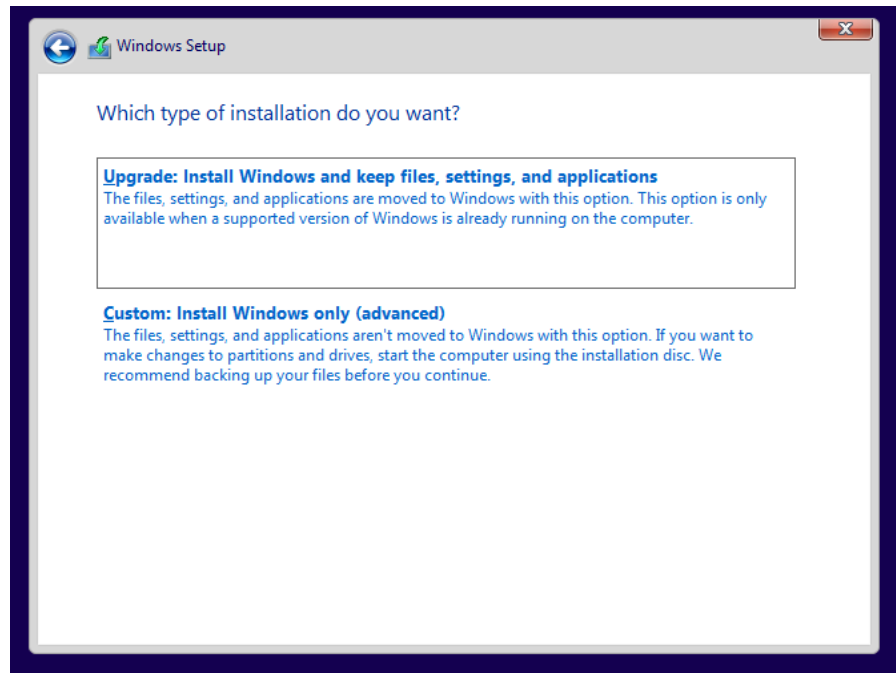
- Select I don't have a product key



- Select **Windows 10 Pro** (Used in real business environments)



- Choose **Custom installation**



- Choose **Set up for personal use**

How would you like to set up?



Set up for personal use

We'll help you set it up with a personal Microsoft account.
You'll have full control over this device.



Set up for an organization

You'll gain access to your organization's resources like email, network, apps, and services. Your organization will have full control over this device.

- Select **Offline account**

Let's add your account

One account connects your device across Microsoft apps and services, like Office, OneDrive, Microsoft Edge, and the Microsoft Store.



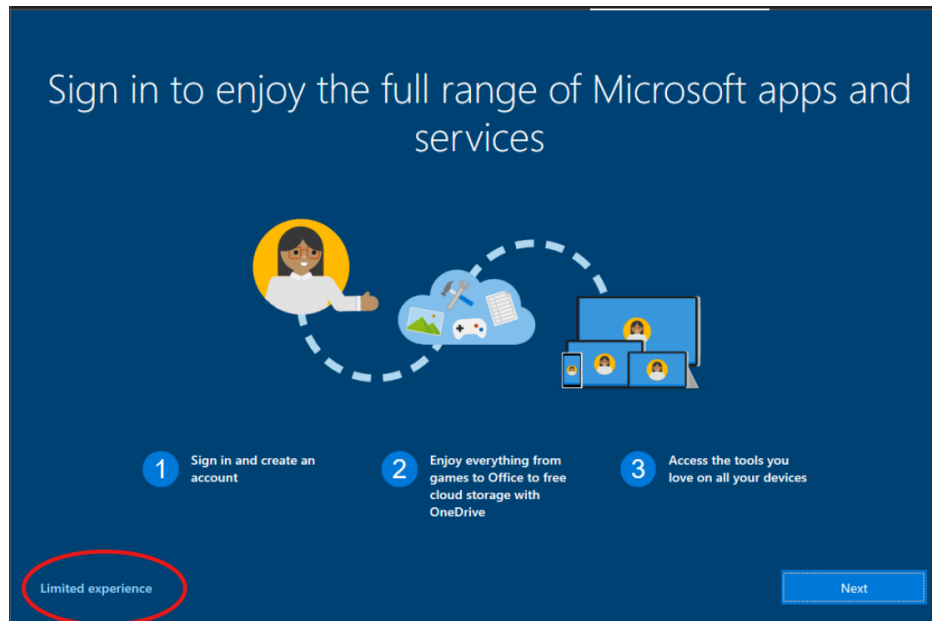
[Create account](#)
[Sign in with a security key](#)

When you sign in with your Microsoft account, we back up your files, including your photos, so you don't have to worry about losing them if something happens to your Windows device. Your settings, files, browsing history, favorites, passwords, contacts, and more, are also synced across devices, apps, and services. See [how your data is managed](#).

[Offline account](#) [Privacy & Cookies](#) [Terms of Use](#)

[Learn More](#) [Next](#)

- Select **Limited experience**

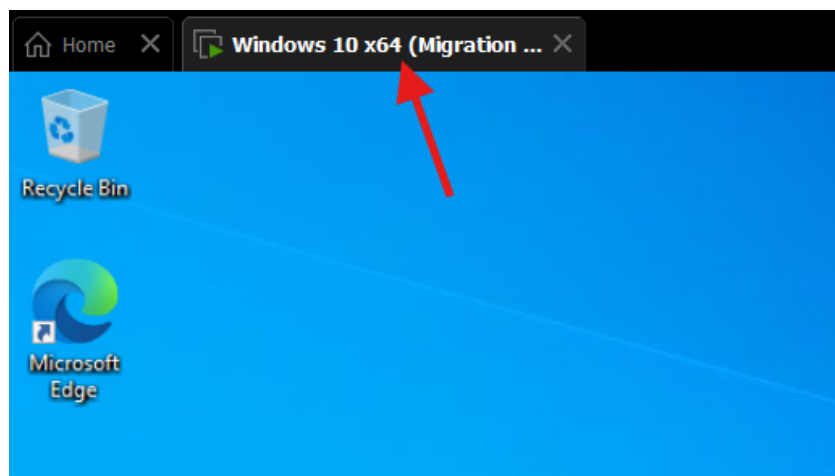


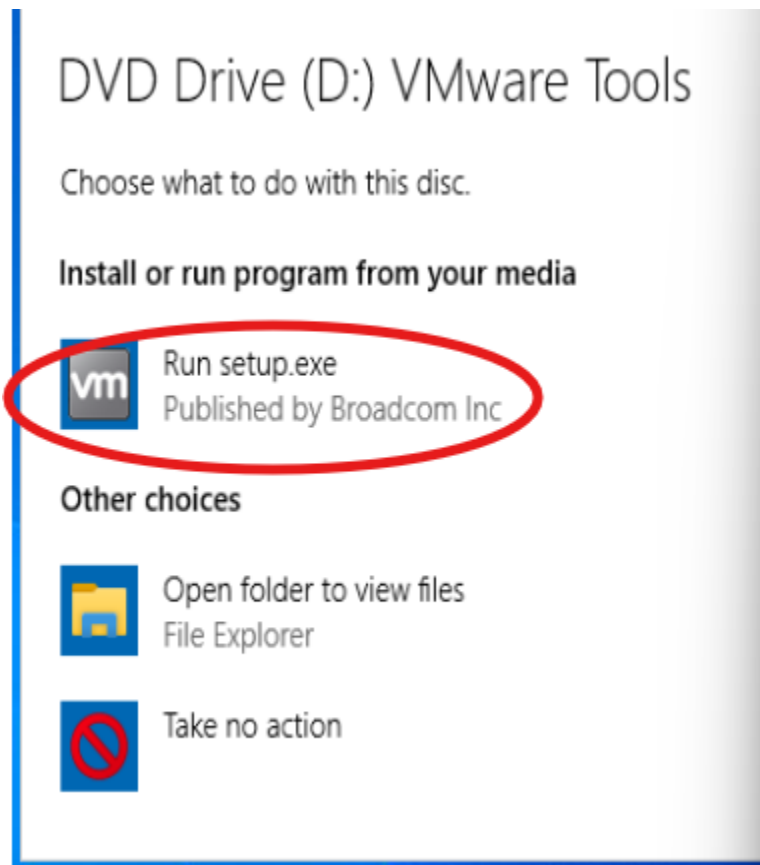
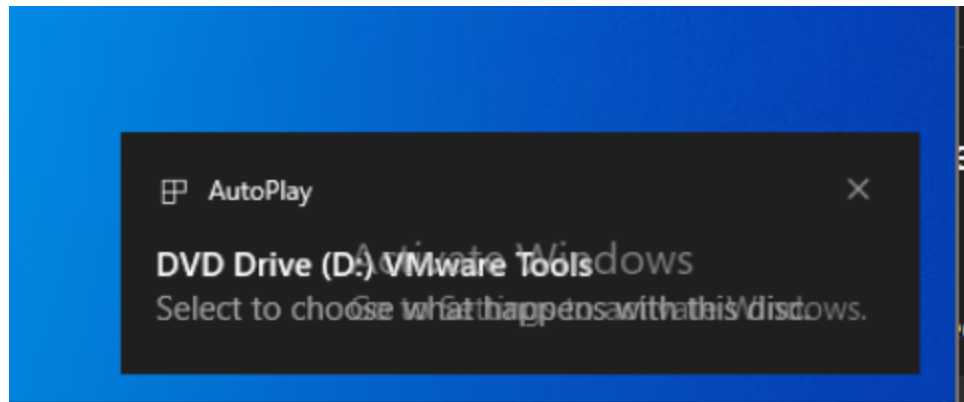
4. The rest of the installation set up is pretty self explanatory.

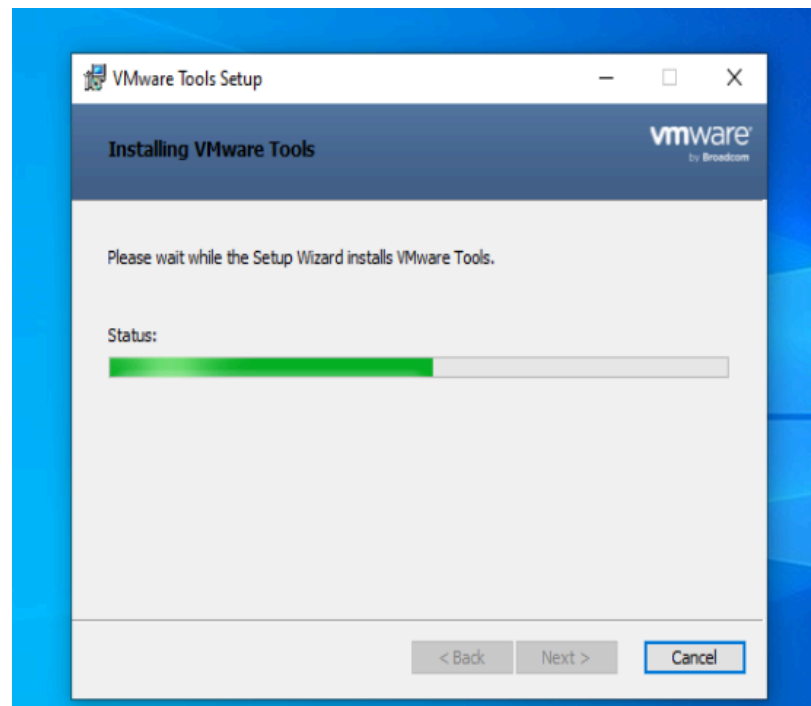
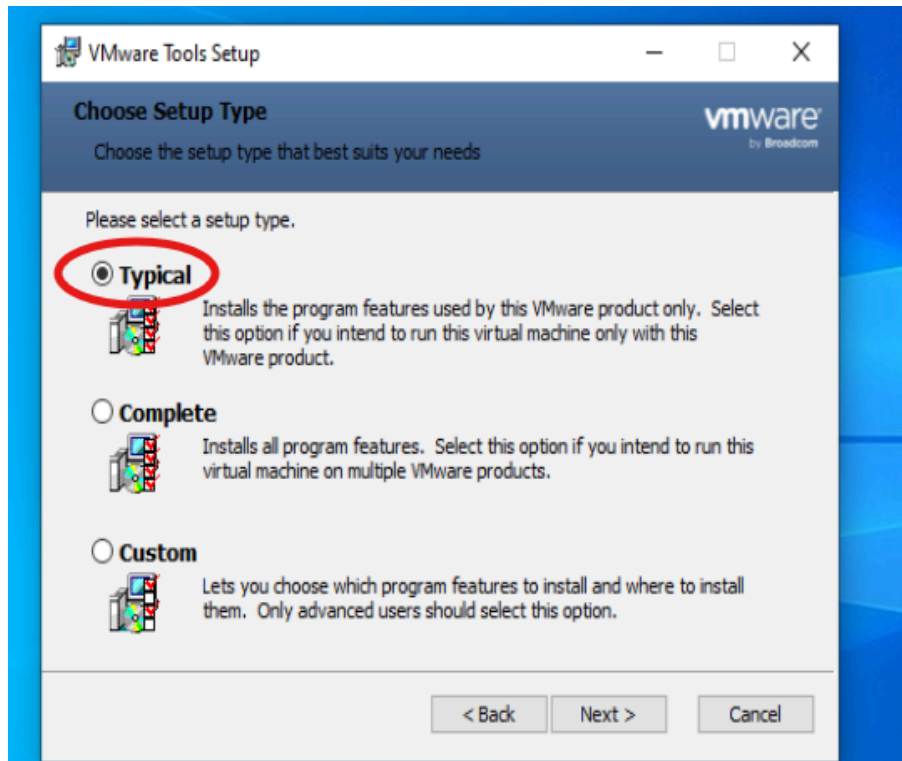
4.2 Post-Install Preparation

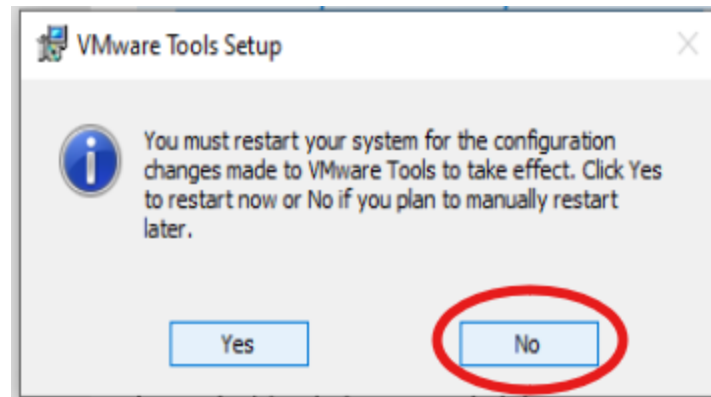
After reaching the Windows 10 desktop:

- Install VMware Tools
 - 1) Right click machine tab at the top and select **Install VM Tools**

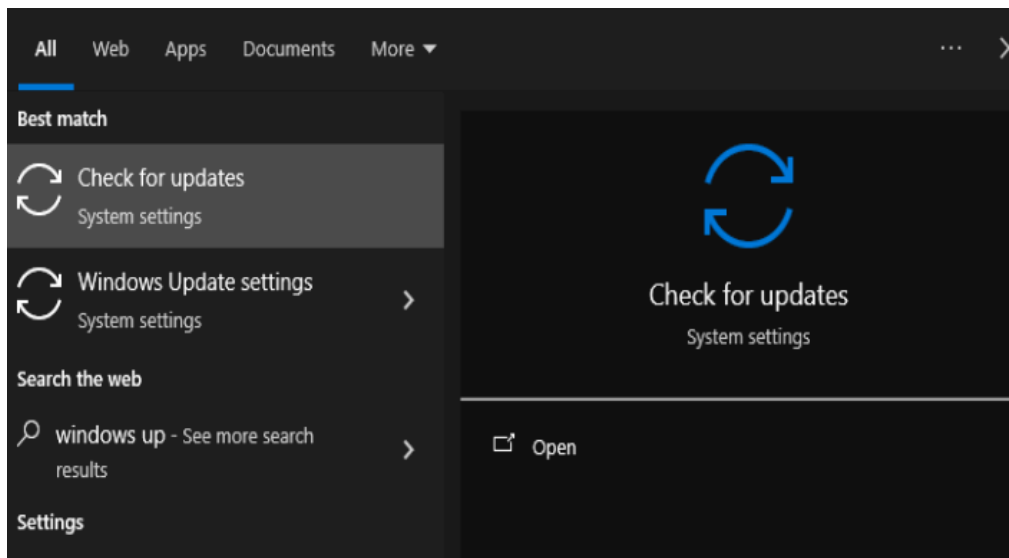








- Run Windows Update until fully patched





Updates available

Last checked: Today, 10:58 PM

Your device is missing important security and quality fixes.

2025-10 Cumulative Update for .NET Framework 3.5, 4.8 and 4.8.1 for Windows 10 Version 22H2 for x64 (KB5066747)

Status: Pending install

2025-10 Cumulative Update for Windows 10 Version 22H2 for x64-based Systems (KB5066791)

Status: Pending install

Microsoft .NET Framework 4.8.1 for Windows 10 Version 22H2 for x64 (KB5011048)

Status: Pending install

[View optional updates](#)



Get the latest updates as soon as they're available



Off

Be among the first to get the latest non-security updates, fixes, and improvements as they roll out. [Learn more](#)



Pause updates for 7 days

Get latest updates to pause again



Install updates as soon as possible

We'll notify you before the restart, and you can postpone if needed



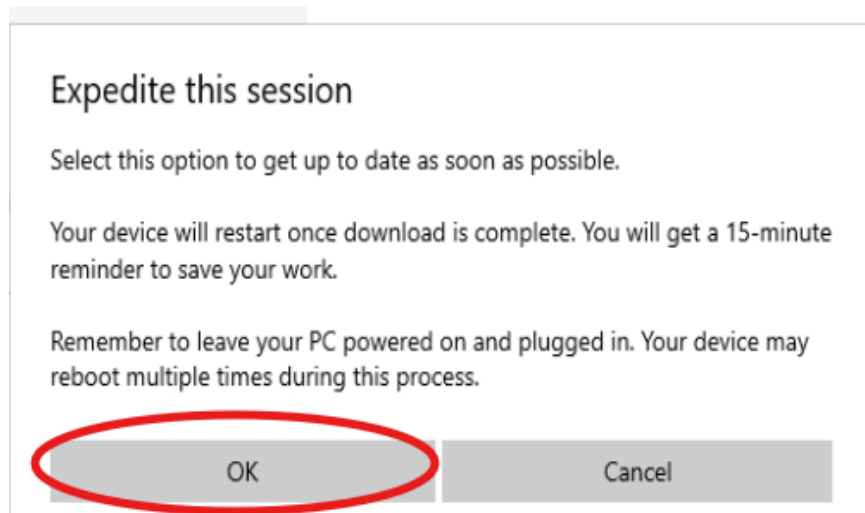
Change active hours

Currently 8:00 AM to 5:00 PM

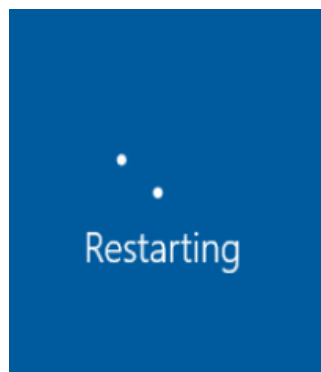
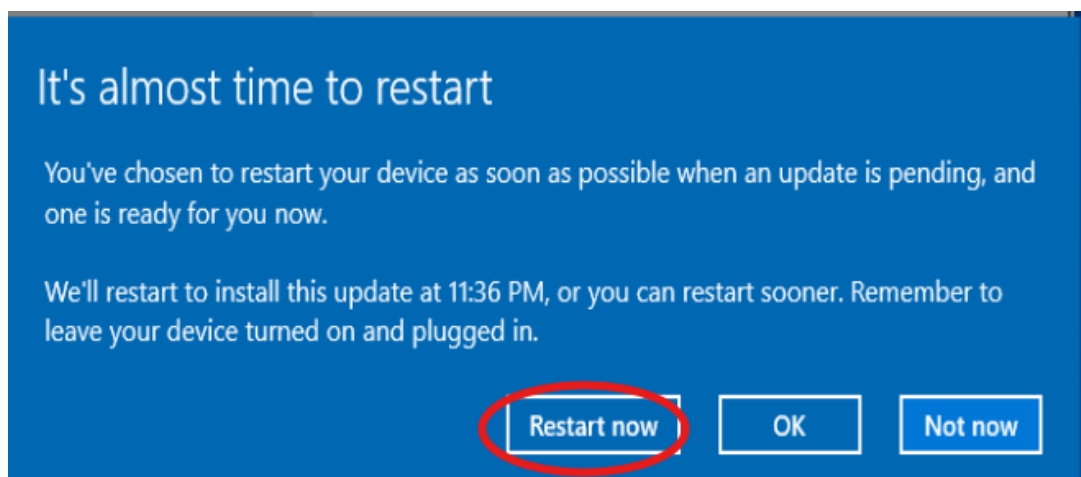


View update history

See updates installed on your device



- Restart as needed

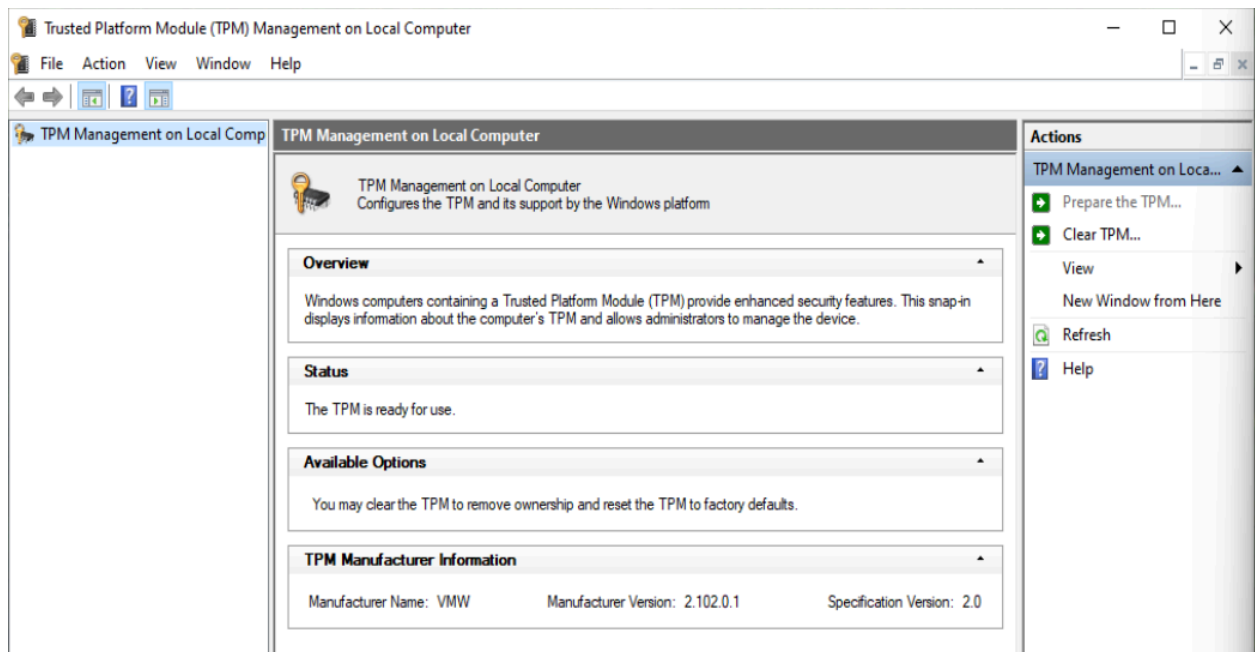
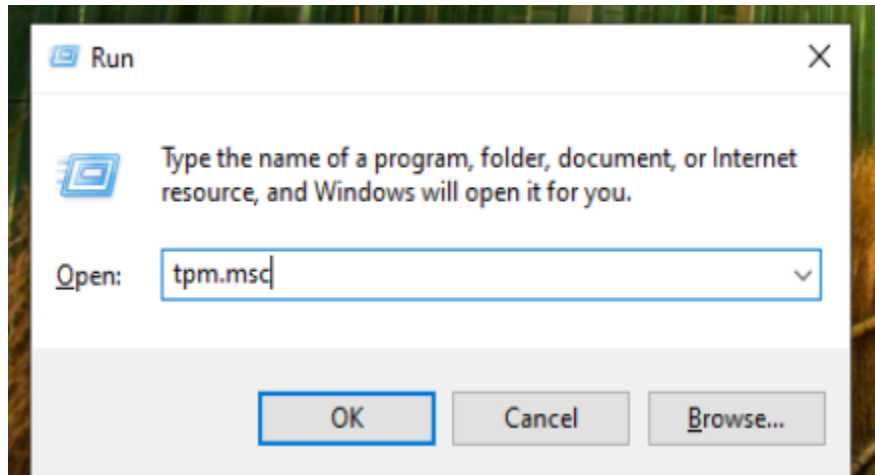


This ensures a stable baseline before migration.

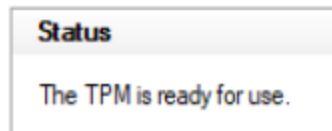
5. Phase 3 – Pre-Migration Readiness Assessment

5.1 Verify TPM 2.0

1. Press **Win + R** → type **tpm.msc**



2. Confirm:
 - Status: *TPM is ready for use*

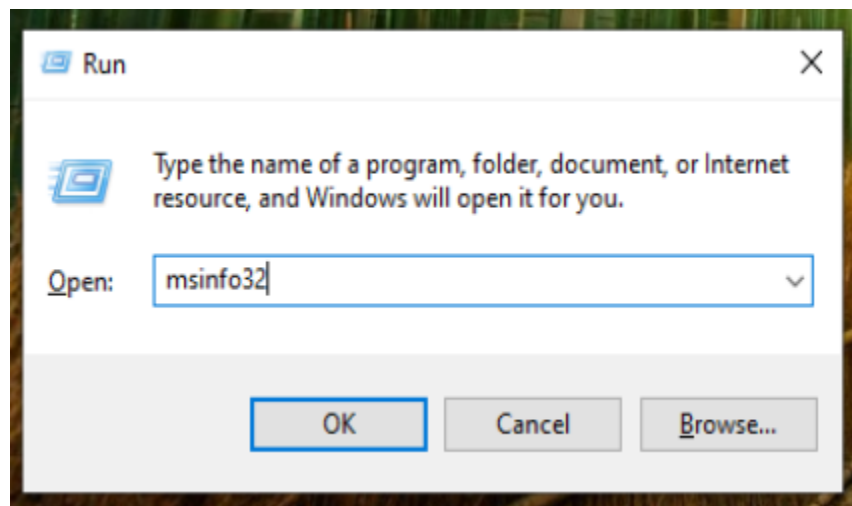


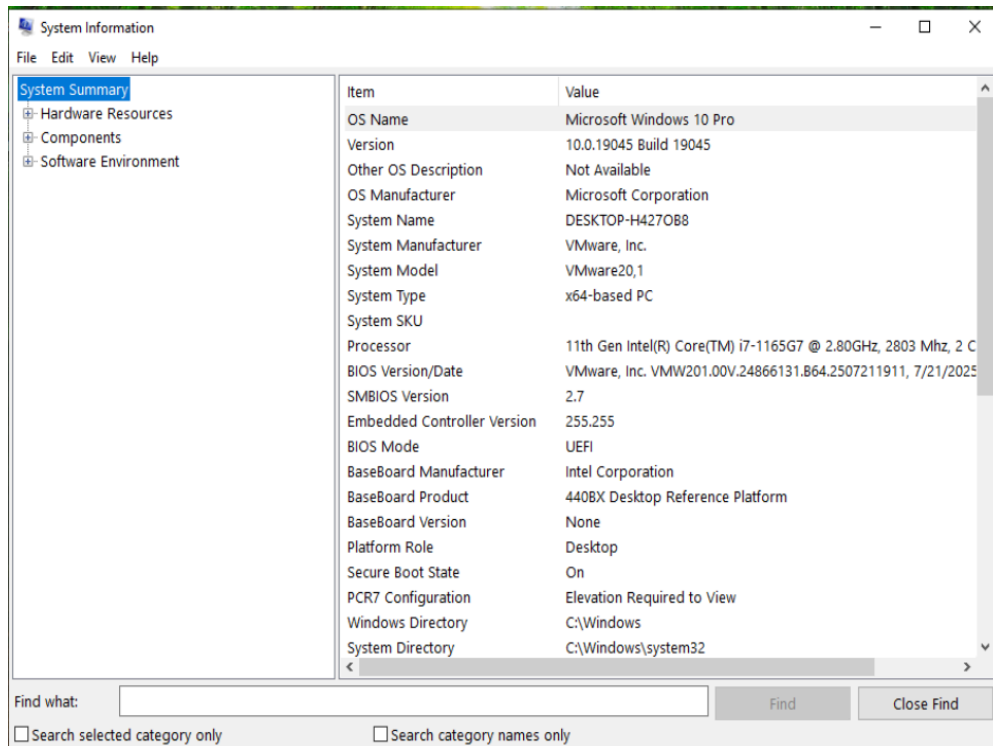
- Specification Version: 2.0



5.2 Verify UEFI & Secure Boot

1. Press **Win + R** → type *msinfo32*





2. Confirm:

- **BIOS Mode: UEFI**

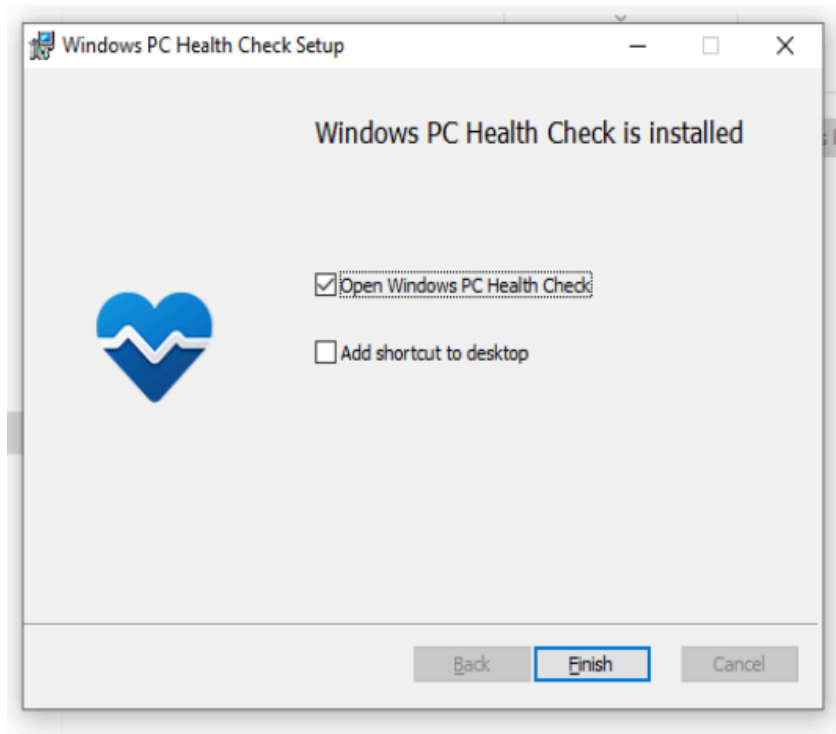
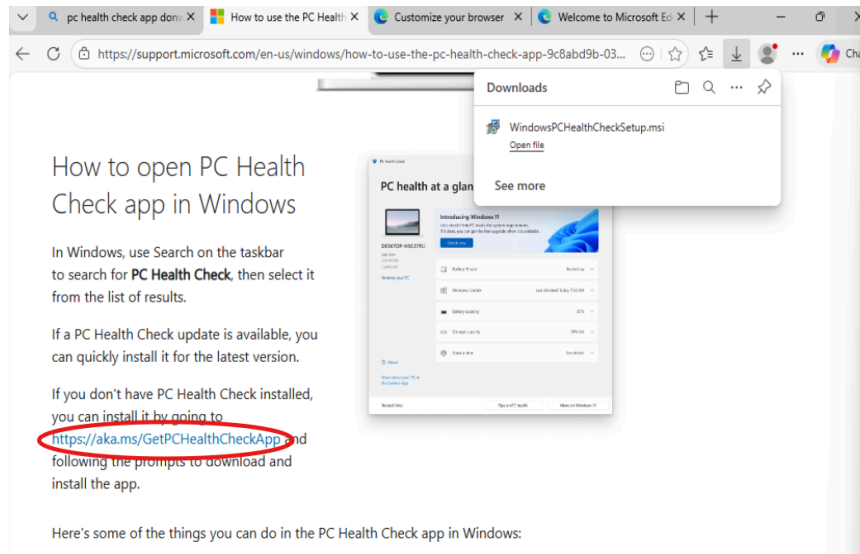
BIOS Mode UEFI

- **Secure Boot State: On**

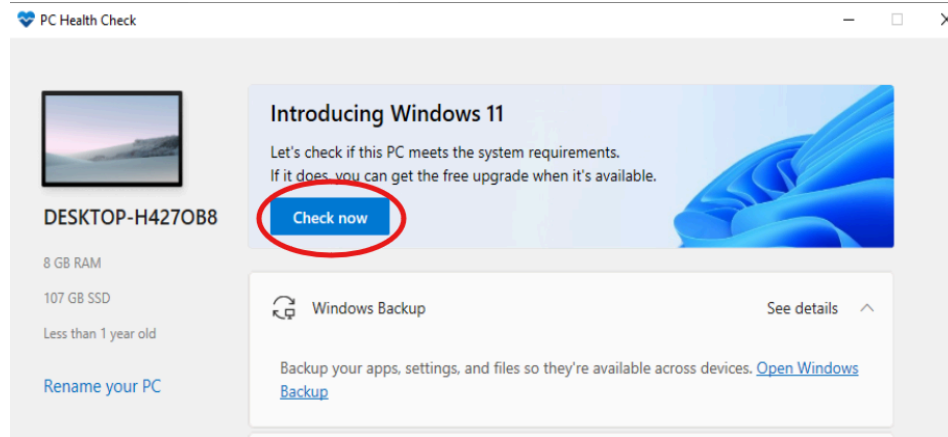
Secure Boot State On

5.3 Windows 11 Compatibility Check

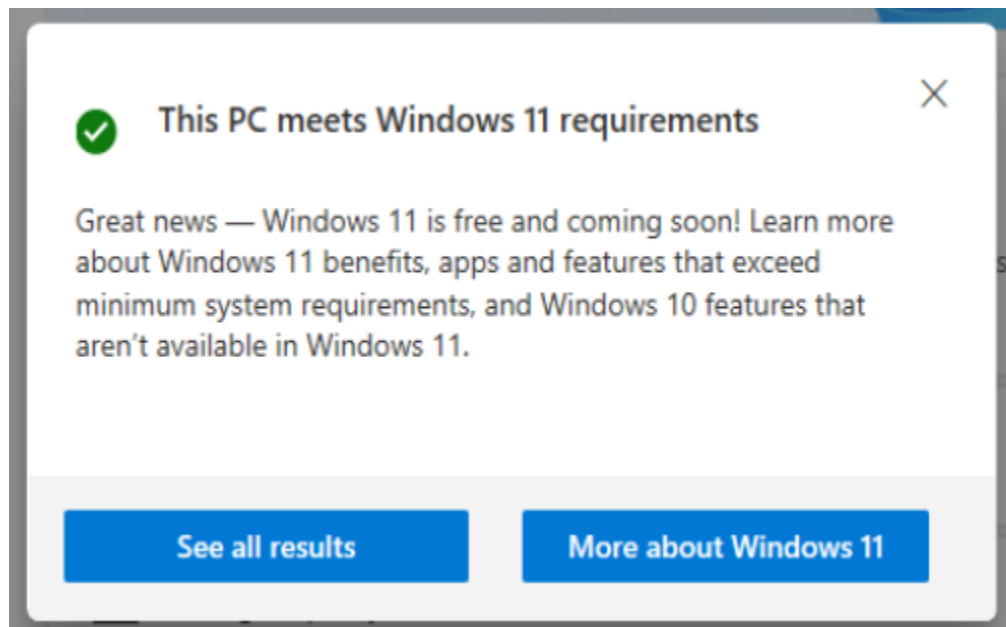
1. Install [PC Health Check](#)



2. Run eligibility assessment



3. Confirm the system is **Windows 11 compatible**

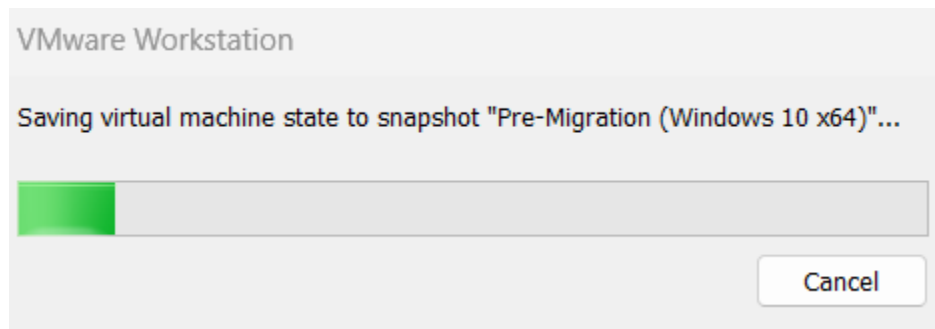
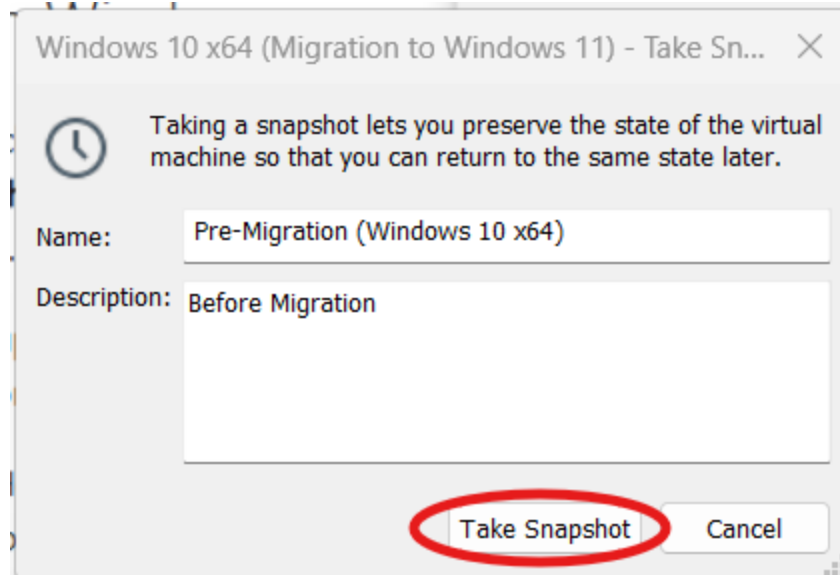


If this fails, stop and correct the configuration (mirrors real IT procedure).

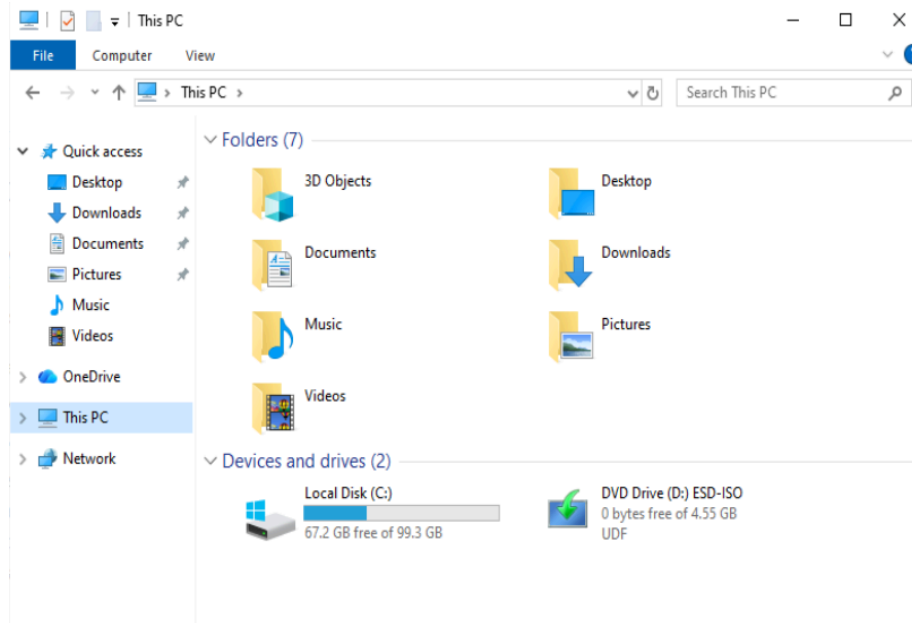
6. Phase 4 – Backup & Rollback Planning

6.1 Backup Strategy (Lab-Appropriate)

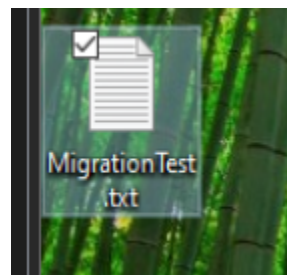
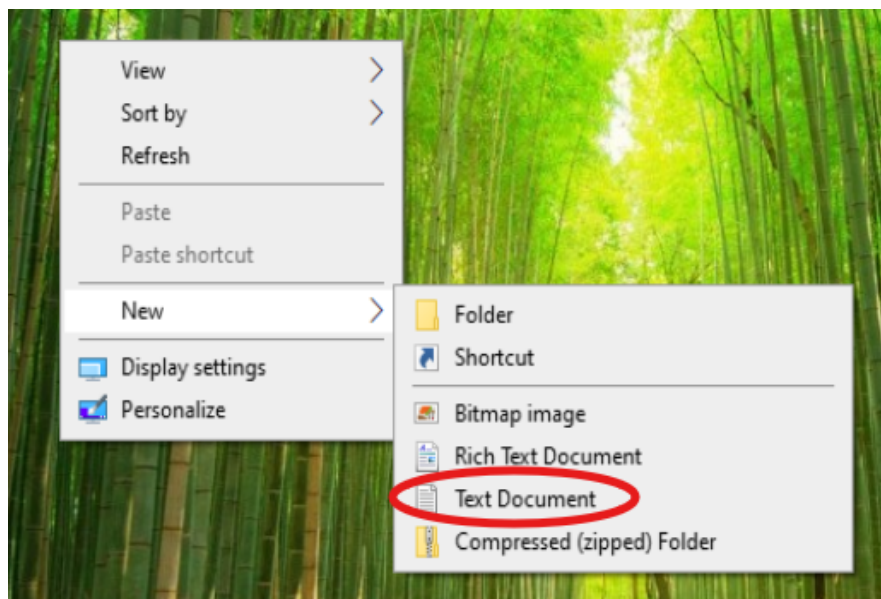
- Create a **VM snapshot**



- Verify user folders exist:
 - Desktop
 - Documents
 - Downloads



- Create a test file, to ensure it is brought along during the migration

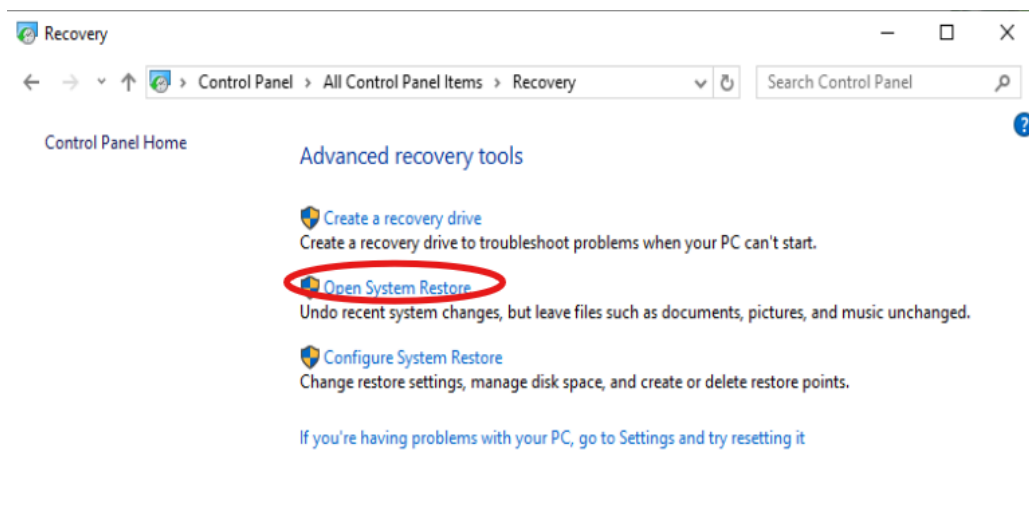


Snapshots simulate enterprise backup and recovery points.

6.2 Rollback Awareness

Windows supports rollback to Windows 10 for approximately **10 days** after upgrade:

Path: Settings → System → Recovery → System Restore



Rollback availability was confirmed before proceeding.

7. Phase 5 – Windows 10 → Windows 11 In-Place Upgrade

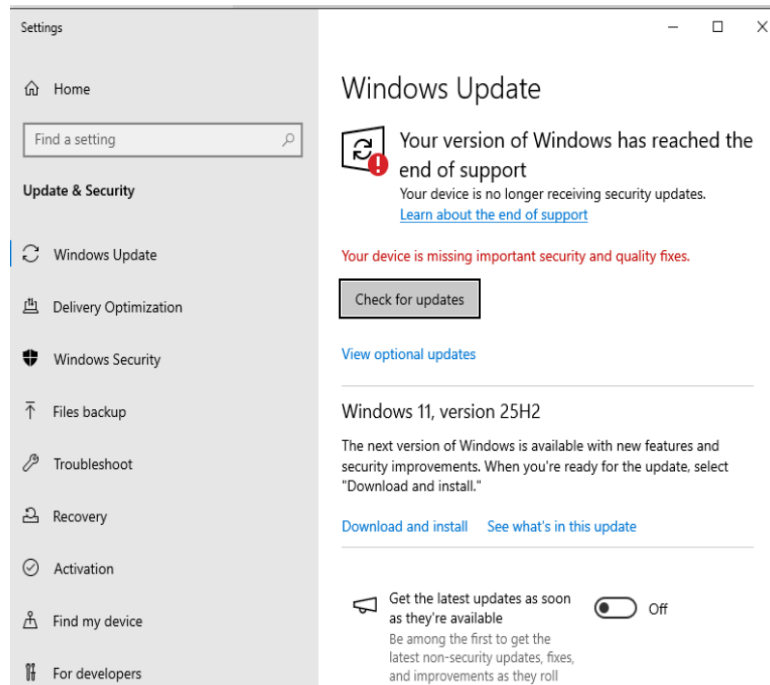
7.1 Upgrade Method Selection

Upgrade Method Used: Windows Update (In-Place Upgrade)

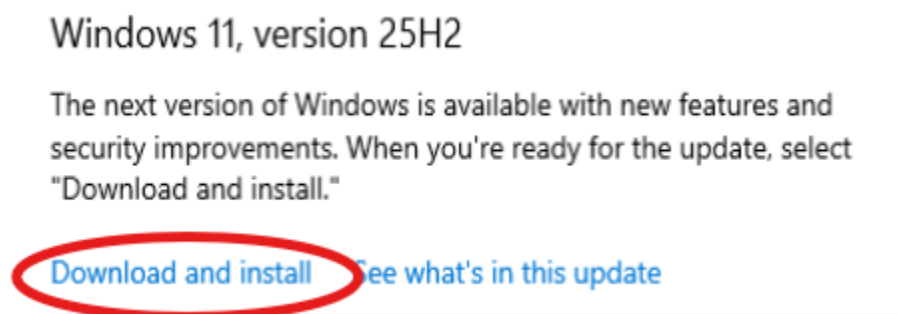
The upgrade can be initiated through Windows Update after the system passes all Windows 11 compatibility checks. This method reflects how pilot users are commonly upgraded in enterprise environments when devices are fully eligible.

7.2 Execute the Upgrade

1. Launch Windows Update



2. Under Windows 11, version 25H2, select Download and install



3. Allow the system to download and install Windows 11

Windows 11, version 25H2
Status: Downloading - 1%

[View optional updates](#)

Windows Update



Restart required

This update is ready to install! We need your help deciding when to restart so we can finish up.

The next Windows feature update is ready and includes reliability, performance, and security improvements.

Windows 11, version 25H2
Status: Pending restart

Restart now

[Schedule the restart](#)

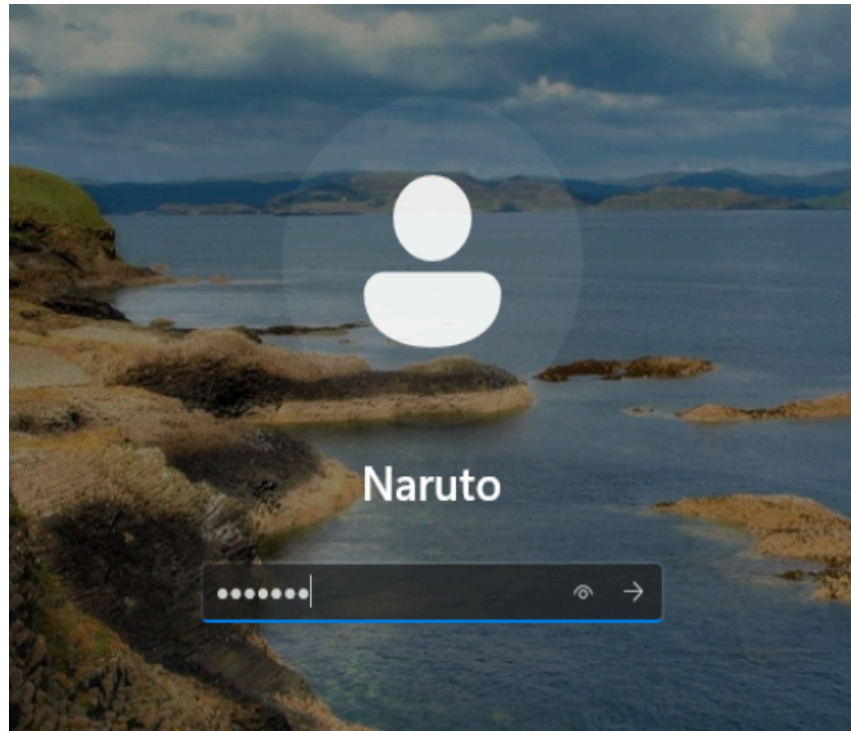
[View optional updates](#)

4. System will reboot multiple times

Do not interrupt the process. It may take a few hours.

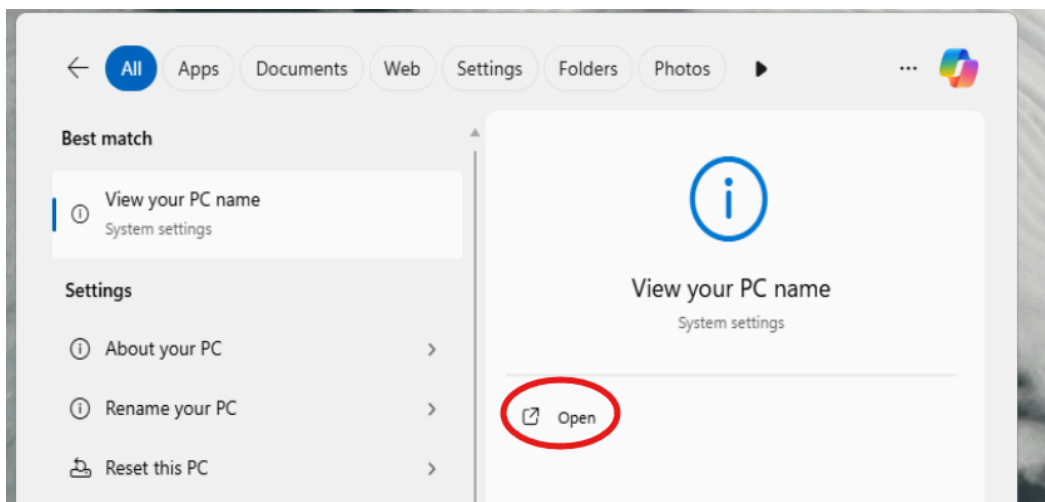
8. Phase 6 – Post-Migration Validation

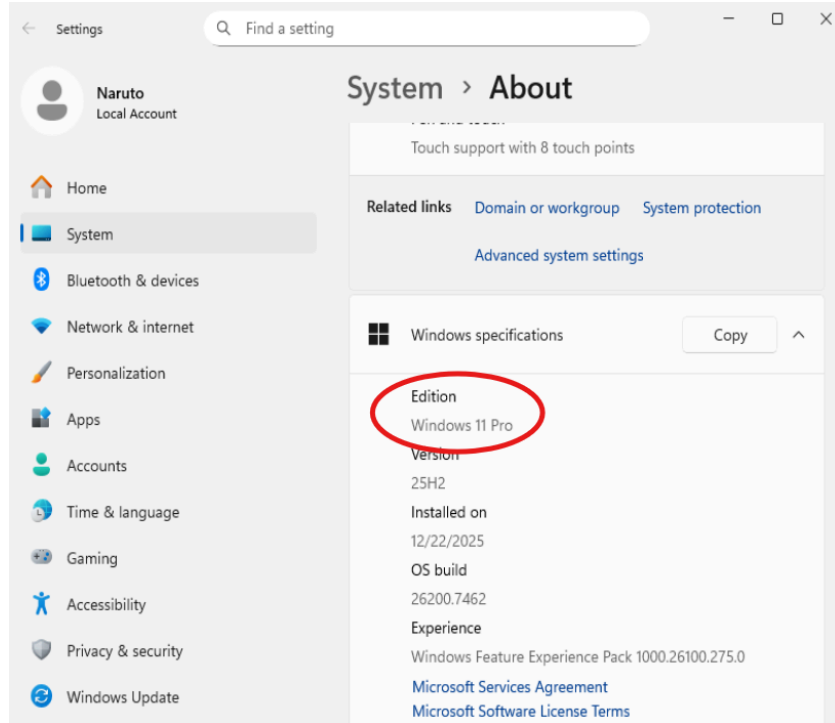
8.1 OS Verification



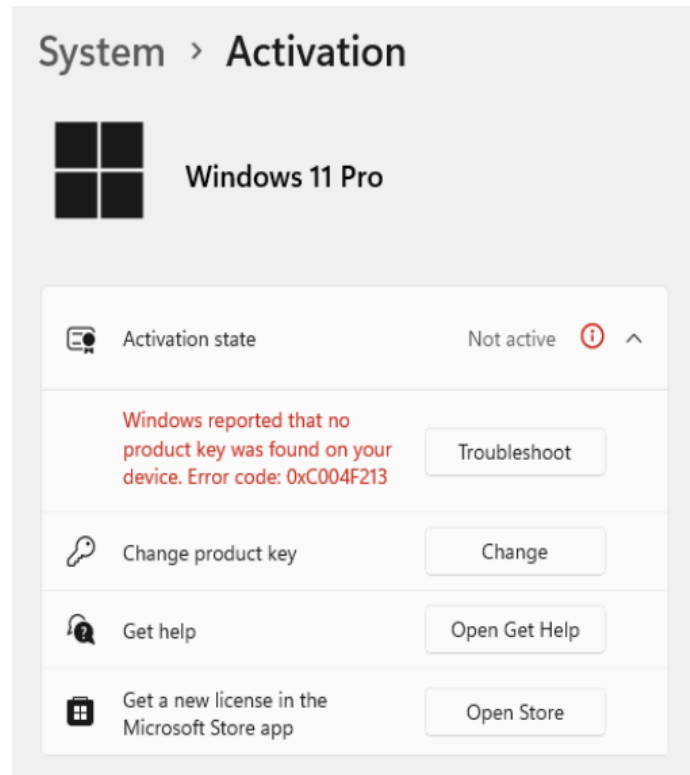
After login:

- Confirm OS: Windows 11



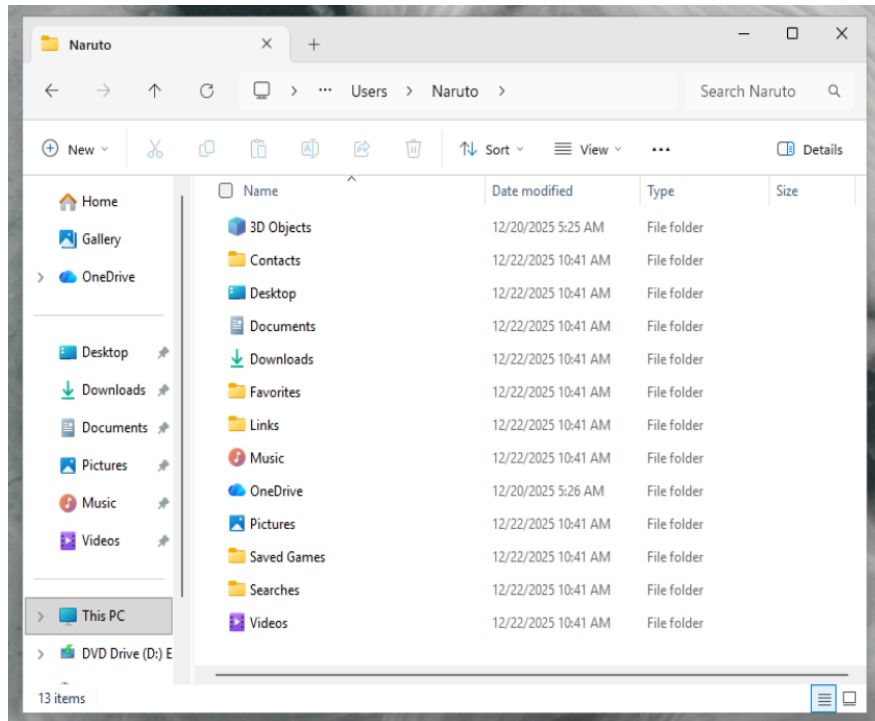


- Check activation status (Windows 11 should have the same activation status as Windows 10 after migration. We did not activate Windows 10 so Windows 11 should not be activated either.)

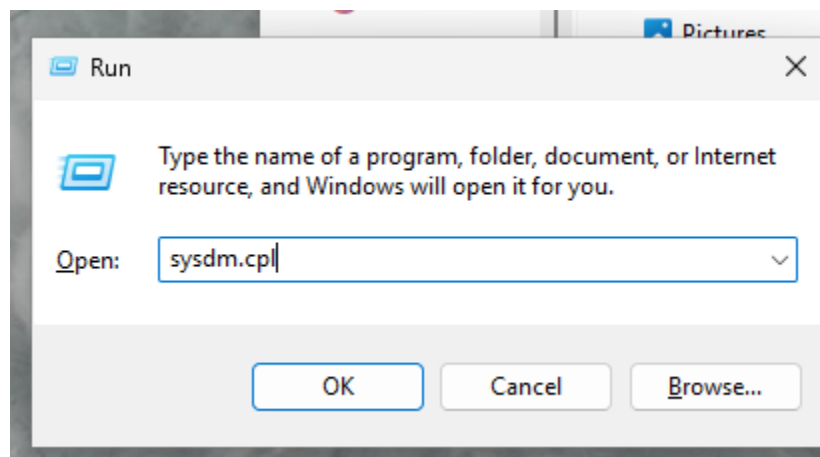


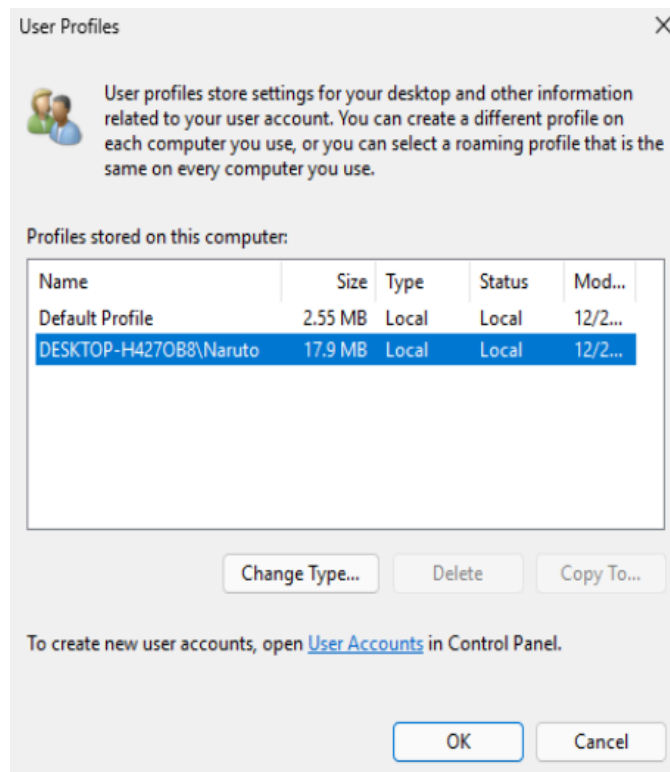
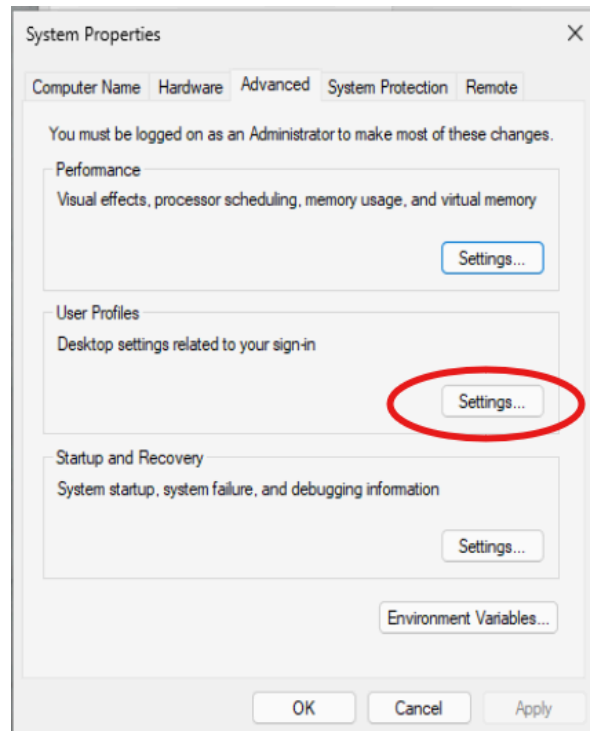
- Verify user profile integrity (Seeing if the user lost anything, making sure all files, folders, etc. are still present.)
 - 1) Check if the test file is still there and check if the user's folders are in their directory



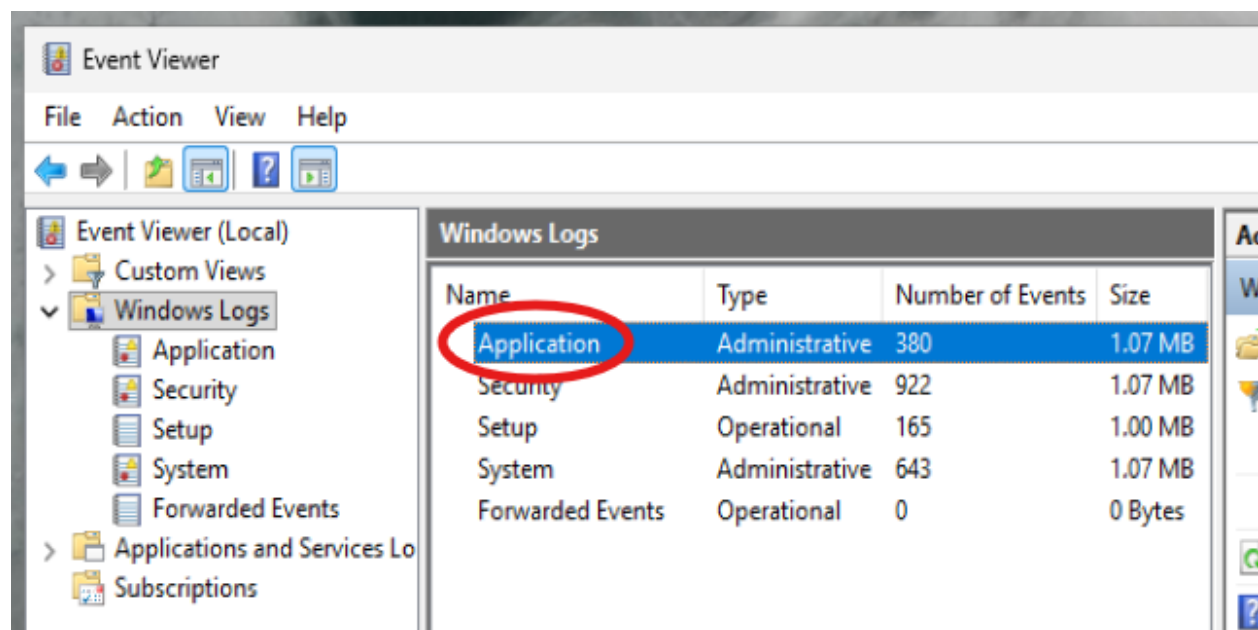
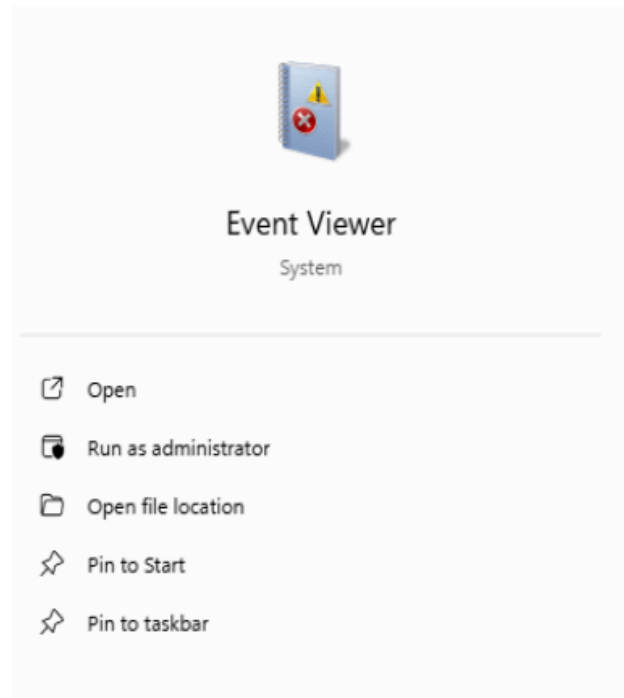


2) Press **Win + R**, and type ***sysdm.cpl*** to check user profile health (check if it's listed and not marked as temporary)





3) Go to **Event Viewer** (Check for User Profile Service errors and profile load failures)



Application Number of events: 380			
Level	Date and Time	Source	Event ID
Information	12/22/2025 10:17:39 PM	MSDTC 2	4202
Error	12/22/2025 10:17:39 PM	Securit...	8198
Information	12/22/2025 10:17:39 PM	Securit...	1003
Information	12/22/2025 10:17:37 PM	Securit...	902
Information	12/22/2025 10:17:37 PM	Securit...	1003
Information	12/22/2025 10:17:37 PM	Securit...	1066
Information	12/22/2025 10:17:37 PM	Securit...	16394
Information	12/22/2025 10:17:37 PM	Securit...	900
Information	12/22/2025 10:17:36 PM	WMI	5617
Information	12/22/2025 10:17:36 PM	VMUp...	271
Information	12/22/2025 10:17:36 PM	VMUp...	250

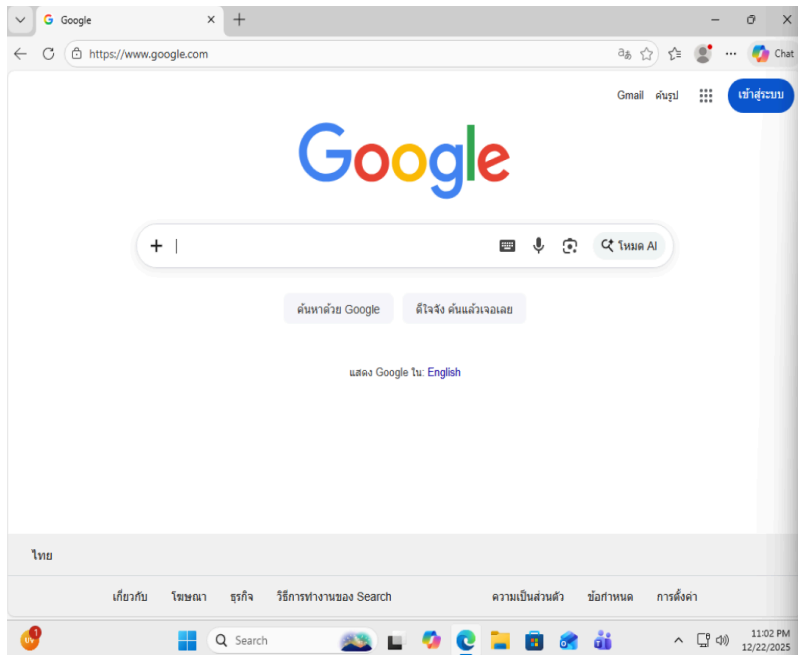
Event 8198, Security-SPP		✕
General Details		
License Activation (slui.exe) failed with the following error code		
hr=0x8007139F		
Command-line arguments:		
RuleId=31e71c49-8da7-4a2f-ad92-45d98a1c79ba;Action=Auto/		

It looks like the only real error is in regards to our Windows License Activation, which is okay because for a VM activation is not necessary.

8.2 Application & System Testing

Validated:

- Internet connectivity



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

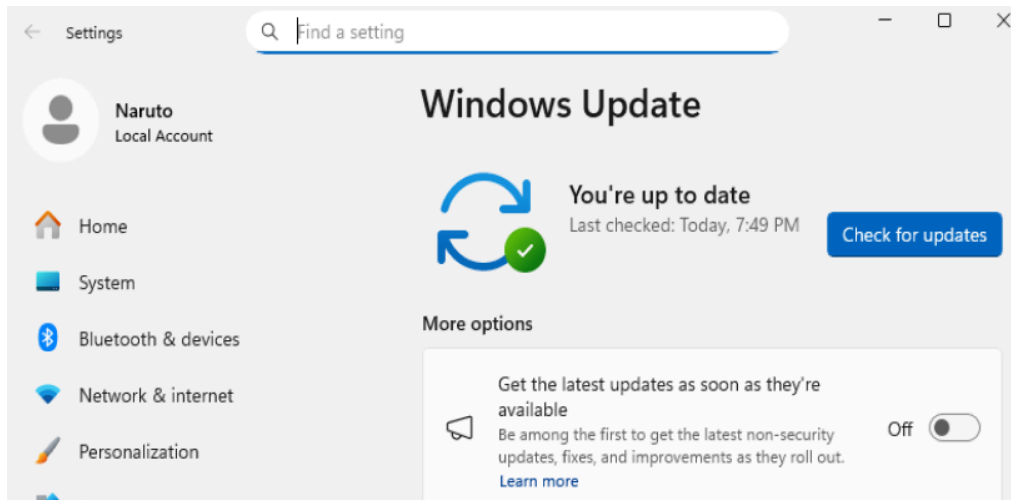
C:\Users\Naruto>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=9ms TTL=128
Reply from 8.8.8.8: bytes=32 time=9ms TTL=128
Reply from 8.8.8.8: bytes=32 time=10ms TTL=128
Reply from 8.8.8.8: bytes=32 time=10ms TTL=128

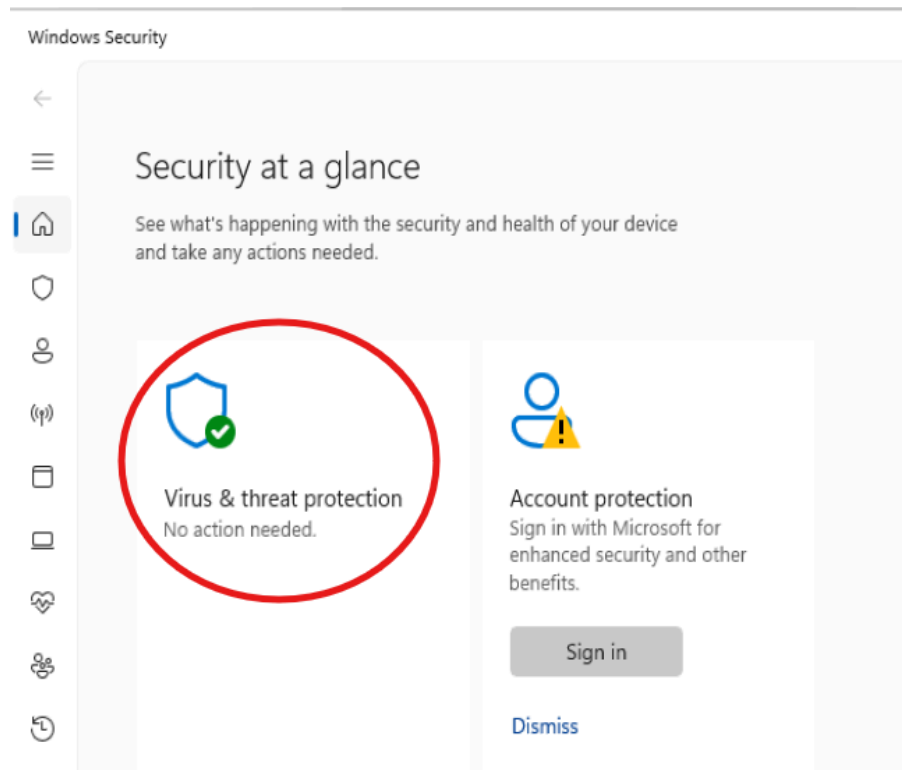
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 10ms, Average = 9ms

C:\Users\Naruto>
```

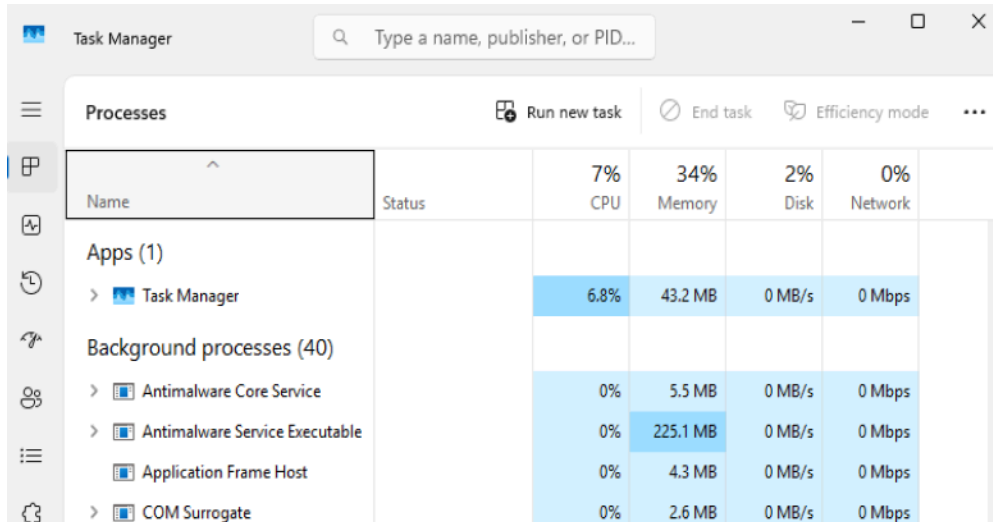
- Windows Update



- Microsoft Defender



- General system performance



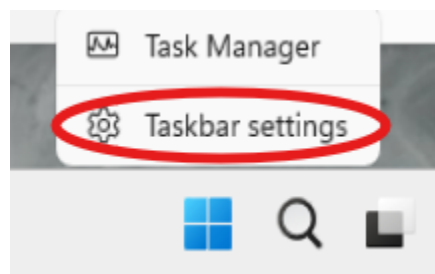
Name	Status	7% CPU	34% Memory	2% Disk	0% Network
Apps (1)					
Task Manager		6.8%	43.2 MB	0 MB/s	0 Mbps
Background processes (40)					
Antimalware Core Service		0%	5.5 MB	0 MB/s	0 Mbps
Antimalware Service Executable		0%	225.1 MB	0 MB/s	0 Mbps
Application Frame Host		0%	4.3 MB	0 MB/s	0 Mbps
COM Surrogate		0%	2.6 MB	0 MB/s	0 Mbps

No critical issues observed.

9. Phase 7 – User Experience & Support Review

9.1 UI Changes Identified

- Centered Start Menu and taskbar (The taskbar on Windows 10 Pro and other earlier versions is on the left by default. Enterprises value consistency, so change the setting to left instead of center of consistency.)



Personalization > Taskbar

Show or hide additional system tray icons

Taskbar behaviors

Taskbar alignment, badging, automatically hide, and multiple displays

Taskbar alignment

Center



Automatically hide the taskbar



Show badges on taskbar apps

Personalization > Taskbar

Show or hide additional system tray icons

Taskbar behaviors

Taskbar alignment, badging, automatically hide,

Left

Taskbar alignment

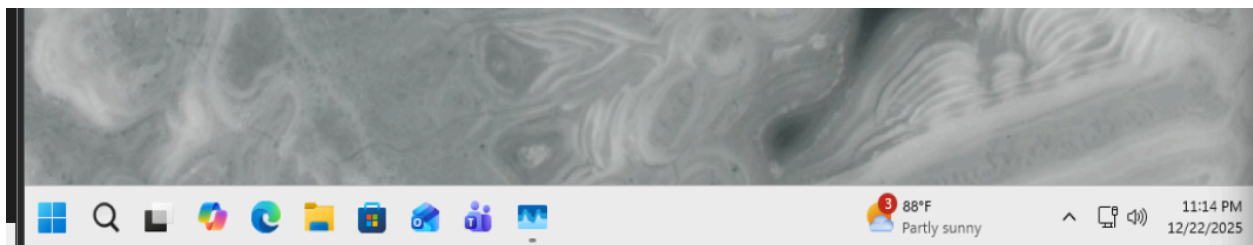
Center



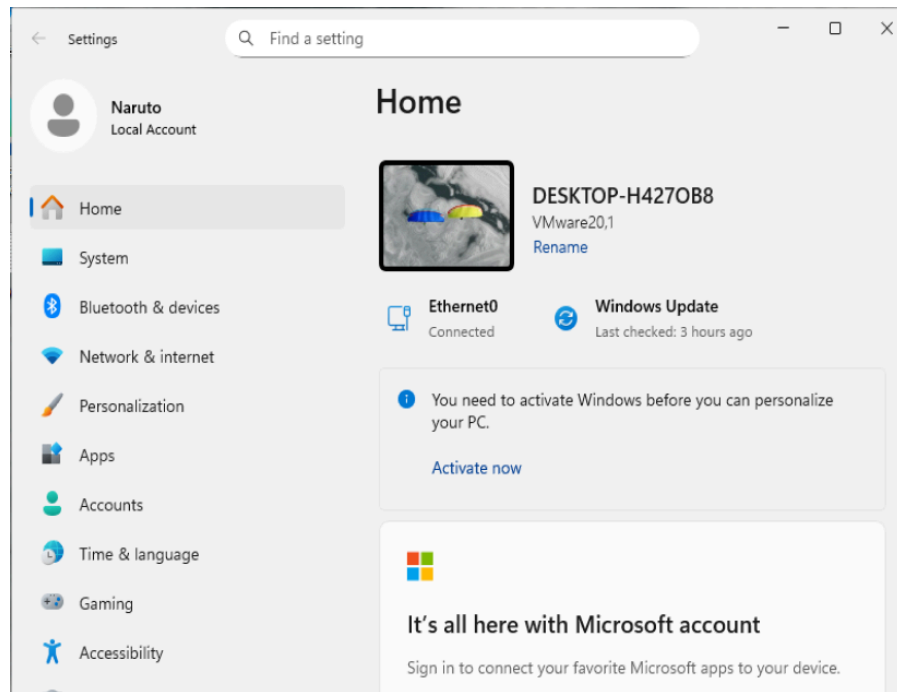
Automatically hide the taskbar



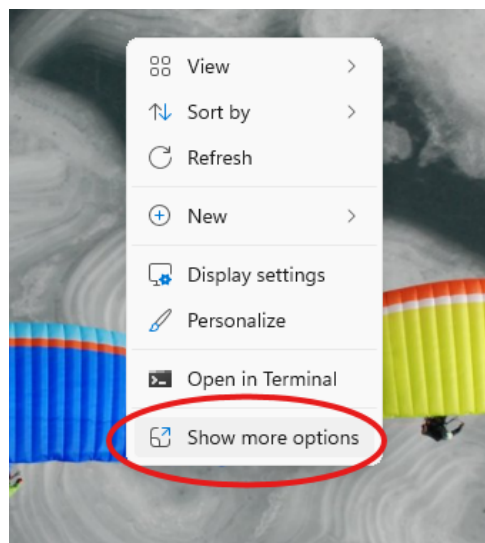
Show badges on taskbar apps

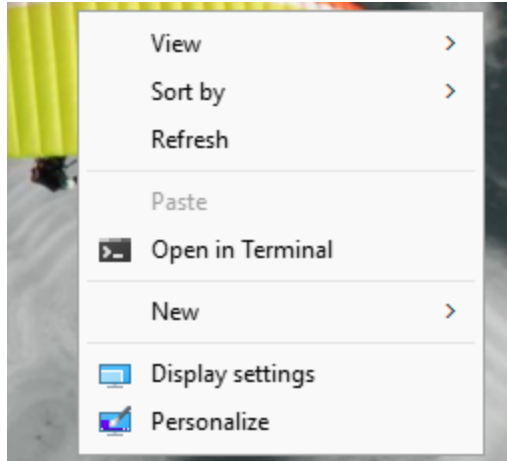


- Updated Settings interface (Look through Settings to see how it's changed, and anticipate updated features that might confuse users.)



- Modified right-click context menu (Verify the Windows 10 options are available and know how to find it. It's a common complaint among Windows 11 users.)





9.2 Anticipated Support Tickets

- “Where are my settings?”
 - Users may be disoriented by the Windows 11 Settings interface and the reduced reliance on the Control Panel
- Taskbar customization requests
 - Users may want to customize their taskbar in a more familiar or comfortable way
- Printer or peripheral issues
 - Due to driver changes, users may face some printer and peripheral device compatibility issues

These are typical post-migration help desk issues.

10. Change Documentation

Change Type: Operating System Feature Upgrade

Risk Level: Moderate

Downtime: Minimal

Upgrade Method: In-place upgrade via Windows Update

Rollback Plan: Enabled (VM snapshot and System Restore configured)

11. Lessons Learned

- Proper VM configuration is critical for Windows 11 eligibility
 - Pre-migration checks prevent upgrade failures
 - In-place upgrades are effective for stable systems
 - Documentation and user readiness are as important as the technical steps
-